What you ignore today will cost you marks tomorrow.

1) Two liquid at different temperatures are mixed in a calorimeter. Which of the following quantities remains conserved?

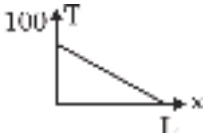
(1) Sum of the temperature of the two liquid.

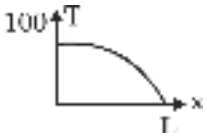
(2) Total heat of the two liquid

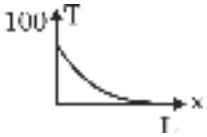
(3) Total internal energy of the two liquid

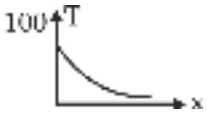
(4) Internal energy of each liquid

2) Figure shows a homogenous rod in shape of a frustum. The temperature vs. position graph will be:

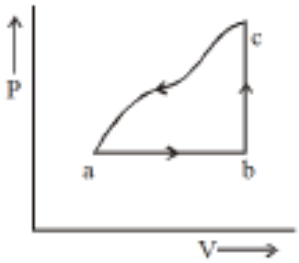
(1) 

(2) 

(3) 

(4) 

3) A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is –180J. The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is :



(1) 100 J

(2) 120 J

(3) 140 J

(4) 130 J

4) If the temperature of the sun were to increase from T to 2T and its radius from R to 2R, then the ratio of the radiant energy received on earth to what it was previously, will be-

(1) 16
(2) 64
(3) 8
(4) 32

5) A container having 1 mol of a gas at a temperature 27°C has movable piston which maintains constant pressure of 1 atm in container. Heat is supplied to the gas until temperature becomes 127°C. The work done by the gas is :-

(1) 703 J
(2) 831 J
(3) 121 J
(4) 2035 J

6) The co-efficient of linear expansion of crystal in one direction is α_1 and in other two perpendicular directions is α_2 then co-efficient of volume expansion will be :-

(1) $2\alpha_1 + \alpha_2$
(2) $3\alpha_1$
(3) $\alpha_1 + 2\alpha_2$
(4) $3\left(\frac{\alpha_1 + \alpha_2}{2}\right)$

7) Two containers of equal volume contains the same gas at a pressure P_1 and P_2 at absolute temperature T_1 and T_2. On Joining the vessel, the gas reaches a common pressure P and common temperature T. The ratio P/T is equal to

(1) $\frac{P_1}{T_1} + \frac{P_2}{T_2}$
(2) $\frac{P_1T_1 + P_2T_2}{(T_1 + T_2)^2}$
(3) $\frac{P_1T_2 + P_2T_1}{(T_1 + T_2)^2}$
(4) $\frac{P_1}{2T_1} + \frac{P_2}{2T_2}$

8) In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation VT = K, where K is a constant. In this process the temperature of the gas is increased by ΔT. The amount of heat absorbed by gas is (R is gas constant) :

(1) $\frac{1}{2}R\Delta T$
(2) $\frac{3}{2}R\Delta T$

(3) $\frac{1}{2}KR\Delta T$

(4) $\frac{2K}{3}\Delta T$

9) Calculate heat required to convert 10 g ice at 0°C into steam at 120°C.

(1) 7200 cal

(2) 7300 cal

(3) 7000 cal

(4) 7400 cal

10) A sample of ideal gas (Degree of freedom f = 5) is heated at constant pressure. If an amount 140 J of heat is supplied to the gas, the work done by the gas is :-

(1) 25 J

(2) 40 J

(3) 35 J

(4) 20 J

11) 1 mole of rigid diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$ the value of x is –

(1) 15

(2) 20

(3) 25

(4) 10

12) If E_m refers to the maximum intensity of radiations and λ_m is the wavelength corresponding to the maximum intensity of radiations, the Wein's displacement law refers to:-

(1) Increase in E_m when temperature is increasing

(2) Decrease in E_m when temperature is increasing

(3) Increase in λ_m when temperature is increasing

(4) Decrease in λ_m when temperature is increasing

13) Equal amount of heat energy are transferred into equal mass of ethyl alcohol and water sample. The rise in temperature of water sample is 25°C. The temperature rise of ethyl alcohol will be (Specific heat of ethyl alcohol is one half of the specific heat of water).

(1) 12.5°C

(2) 25°C

(3) 50°C

(4) It depends on the rate of energy transfer

14) For a given gas at 1 atm pressure, rms speed of the molecule is 200 m/s at 127°C. At 2 atm pressure and at 227°C, the rms speed of the molecules will be :

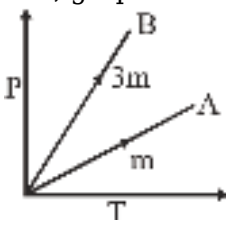
(1) 80 m/s

(2) $100\sqrt{5}$ m/s

(3) $80\sqrt{5}$ m/s

(4) 100 m/s

15) Two different masses m and 3m of an ideal gas are heated separately in a vessel of constant volume, the pressure P and absolute temperature T, graph for these two cases are shown in figure as A and B. The ratio of slopes of curve B to A is :-



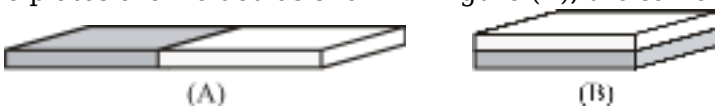
(1) 3 : 1

(2) 1 : 3

(3) 9 : 1

(4) 1 : 9

16) Two identical plates of metal are welded end to end as shown in figure (A); 20 cal of heat flows through it in 4 minutes. If the plates are welded as shown in figure (B), the same amount of heat will flow through the plates in :-



(1) 1 minute

(2) 2 minute

(3) 4 minute

(4) 16 minute

17) 500 g of water and 100 g of ice at 0°C are in a calorimeter whose water equivalent is 40 g. 10g of steam at 100°C is added to it. Then water in the calorimeter is (in gm) : (Latent heat of ice = 80 cal/g, Latent heat of steam = 540 cal/g)

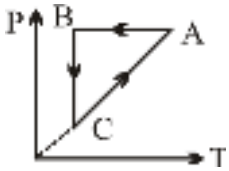
(1) 600 gm

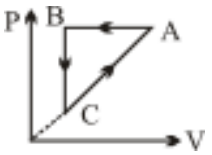
(2) 650 gm

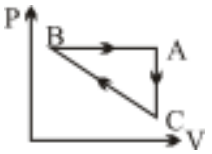
(3) 590 gm

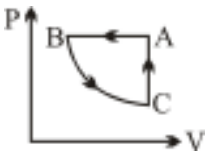
(4) 500 gm

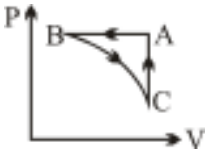
18) A cyclic process ABCA is shown in P-T diagram. When presented on PV, it would



(1) 

(2) 

(3) 

(4) 

19) A particle performs Linear SHM such that it is placed on a platform & platform along with particle oscillate vertically up and down with amplitude A = 1 cm. If the particle do not loose contact with platform any where and mass of particle is 1 kg find.
The minimum possible time period (take $\pi = \sqrt{g}$)

(1) 0.1 sec
(2) 0.2 sec
(3) 0.3 sec
(4) 0.4 sec

20) Two simple harmonic waves travelling in the same direction with same amplitude and frequency are superimposed. The resultant amplitude is equal to the $\sqrt{3}$ times of amplitude of individual motions. The phase difference between the two motions is _____

(1) $\frac{\pi}{6}$
(2) $\frac{\pi}{3}$
(3) $\frac{2\pi}{3}$
(4) $\frac{4\pi}{3}$

21) Two particles perform SHM with the same amplitude and same frequency about the same mean position and along the same line. If the maximum distance between them during the motion is A (A is the amplitude of either) then the phase difference between their SHM is :

(1) $\frac{\pi}{6}$
(2) $\frac{\pi}{3}$
(3) $\frac{\pi}{2}$

(4) $\frac{2\pi}{3}$

22) A particle undergoing simple harmonic motion has time dependent displacement given by $x(t) = A\sin\frac{\pi t}{90}$. The ratio of kinetic to potential energy of this particle at t = 210 s will be :

(1) 2
(2) 3
(3) $\frac{1}{3}$
(4) 1

23) A particle executes simple harmonic motion with an amplitude of 5 cm. When the particle is at 4 cm from the mean position, the magnitude of its velocity in SI units is equal to magnitude of its acceleration. Then, its periodic time in seconds is :

(1) $\frac{7}{3}\pi$
(2) $\frac{3}{8}\pi$
(3) $\frac{4\pi}{3}$
(4) $\frac{8\pi}{3}$

24) The displacement of a particle varies according to the relation $x = 3\sin 100t + 8\cos^2 50t$. Which of the following is INCORRECT about this motion

(1) The motion of the particle is S.H.M.
(2) The amplitude of the S.H.M. of the particle is 5 units
(3) The amplitude of the resultant S.H. M. is $\sqrt{73}$ units
(4) The maximum displacement of the particle from the origin is 9 units .

25) 2 particles A & B of equal masses are suspended from two massless springs of constant k_1 & k_2 respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A & B is :-

(1) $\frac{k_2}{k_1}$
(2) $\frac{k_1}{k_2}$
(3) $\sqrt{\frac{k_1}{k_2}}$
(4) $\sqrt{\frac{k_2}{k_1}}$

26) A wall clock uses a vertical spring-mass system to measure the time. Each time the mass reaches

an extreme position, the clock advances by a second. The clock gives correct time at the equator. If the clock is taken to the poles it will : (neglect thermal effect)

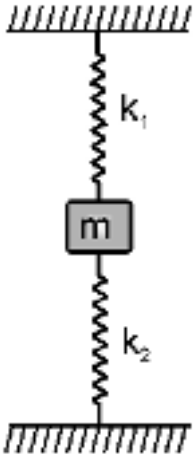
(1) run slow.

(2) run fast.

(3) stop working.

(4) give correct time.

27) In the figure shown, a block of mass m is attached at ends of two springs. The other ends of the spring are fixed. The mass m is released in the vertical plane when the spring are relaxed. The velocity of the block is maximum when:



(1) k_1 is compressed and k_2 is elongated

(2) k_1 is elongated and k_2 is compressed

(3) k_1 and k_2 both are compressed

(4) k_1 and k_2 both are elongated.

28) **Statement-I :** Position–time equation of a particle moving along x–axis is $x = 4 + 6 \sin\omega t$. Under this situation, motion of particle is not simple harmonic.

Statement-II : $\frac{d^2x}{dt^2}$ for the given equation is not proportional to –x.

(1) Statement–I is True, Statement–II is True ; Statement–II is a correct explanation for Statement–I.

(2) Statement–I is True, Statement–II is True ; Statement–II is not a correct explanation for Statement–I.

(3) Statement–I is True, Statement–II is False.

(4) Statement–I is False, Statement–II is True.

29) Vibrations of a string of length 100 cm fixed at both ends are represented by the equation $y = 8\sin\left(\frac{\pi x}{20}\right)\cos(50\pi t)$ Where x and y are in cm and t in seconds.
Maximum displacement at x = 5 cm is :

(1) 5.6 cm

(2) 7.2 cm

(3) 4.4 cm

(4) 3.8 cm

30) The equation of a plane progressive wave is

$y = 0.03 \sin 8\pi\left[t - \frac{x}{20}\right]$. When it is reflected at a rarer medium (medium with higher velocity) at x = 0, its amplitude becomes 50% of its previous value. The equation of the reflected wave is

(1) $y = 0.02 \sin 8\pi \left[t - \frac{x}{20}\right]$

(2) $y = 0.02 \sin 8\pi \left[t + \frac{x}{20}\right]$

(3) $y = +0.015 \sin 8\pi \left[t + \frac{x}{20}\right]$

(4) $y = -0.015 \sin 8\pi \left[t + \frac{x}{20}\right]$

31) A person speaking normally produces a sound intensity of 40 dB at a distance of 1 m. If the threshold intensity for reasonable audibility is 20 dB, the maximum distance at which person can be heard clearly is :

(1) 4 m

(2) 5 m

(3) 10 m

(4) 20 m

32) A unknown source of sound gives 5 beats per second when sounded with another source of frequency 100 Hz. The third harmonic of the unknown source, together with a source of frequency 310 Hz gives 5 beats per second. What is the frequency of the source ?

(1) 105 Hz

(2) 95 Hz

(3) 315 Hz

(4) 305 Hz

33) Two open organ pipes of length 50 cm and 50.5 cm produce 3 beats/sec when sounded together in fundamental modes. The velocity of sound is -

(1) 303 m/s

(2) 330 m/s

(3) 151.5 m/s

(4) 603 m/s

34) Two source of sound placed close to each other are emitting progressive wave given by
$y_1 = 4\sin 600\pi t$
$y_2 = 5\sin 608\pi t$
match the column

	Column-I		**Column-II**

i	Beat frequency	a	302 Hz
ii	frequency of amplitude variation	b	2Hz
iii	frequency of resulting wave	c	4Hz
iv	frequency of intensity variation	d	300Hz

(1) (i)→b, (ii)→c, (iii)→a, (iv)→c

(2) (i)→c, (ii)→b, (iii)→(a), (iv)→c

(3) (i)→c, (ii)→b, (iii)→a, (iv)→d

(4) None of these

35) The velocity of sound in hydrogen at 0°C is 1200 m/s. When some amount of oxygen is mixed with hydrogen, the velocity decreases to 500 m/s. The ratio of volume of H_2 to O_2 in this mixture, (given that $\rho_{O_2} = 16\rho_{H_2}$)

(1) 119/256

(2) 256/119

(3) 12/5

(4) None of these

36) A hollow pipe of length 0.8 m is closed at one end. At its open end a 0.5 m long uniform string is vibrating in its second harmonic and it resonates with the fundamental frequency of the pipe. If the tension in the wire is 50 N and the speed of sound is 320 ms^{-1}, the mass of the string is :

(1) 5 grams

(2) 10 grams

(3) 20 grams

(4) 40 grams

37) In a closed organ pipe, the frequency of fundamental note is 30 Hz. A certain amount of water is now poured in the organ pipe so that the fundamental frequency is increased to 33 Hz. If the organ pipe has a cross-sectional area of 3 cm^2, the amount of water poured in the organ tube in _____ g. (Take speed of sound in air is 330 m/s)

(1) 25

(2) 10

(3) 75

(4) 50

38) **Assertion** : The fundamental frequency of an open organ pipe increases as the temperature is increased.
Reason : As the temperature increases, the velocity of sound increases more rapidly than the length of the pipe.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

39) A sonometer wire resonate with T.F. forming standing wave with five antinode between the two bridges when a mass of 16 kg is suspended from the wire. When the mass is replaced by mass M the wire resonate with same T.F. forming two antinode for the same position of bridges. Find out the value of M.

(1) 10 kg

(2) 100 kg

(3) 4 kg

(4) 25 kg

40) Length of a resonance tube is 130 cm. Speed of sound is 320 m/s and frequency of the source is 320 Hz.

	Column-I		**Column-II**
(a)	At resonance, minimum length of air column (in cm)	(i)	105
(b)	At resonance, minimum length of water column (in cm)	(ii)	3
(c)	At resonance, maximum length of water column (in cm)	(iii)	125
(d)	At resonance, maximum length of air column (in cm)	(iv)	25
(e)	Total number of resonance obtained is	(v)	5

Correct match between the columns is :

(1) a-(iv); b-(v); c-(i); d-(iii); e-(v)

(2) a-(iv); b-(i); c-(i); d-(iii); e-(ii)

(3) a-(iv); b-(v); c-(i); d-(iii); e-(ii)

(4) a-(iv); b-(iii); c-(i); d-(iv); e-(ii)

41) A pendulum with length 1.0 m is released from an initial angle of 8° from vertical. After 1000 s, its amplitude has been reduced to 4°. After how much time its amplitude is reduced to 1°.

(1) 18 min

(2) 30 min

(3) 45 min

(4) 50 min

42) A solid sphere of mass 5kg at 80°C is placed gently in a liquid of mass 25kg at 16°C and of specific heat 1 cal/g°C. At thermal equilibrium, the temperature of system is 40°C, the specific heat of material of sphere is (Neglect heat capacity of vessel).

(1) 2 cal/g°C

(2) 3 cal/g°C

(3) 4 cal/g°C

(4) 6 cal/g°C

43) Three different liquids A, B and C have same mass but having different temperature 10°C, 15°C and 25°C. When A and B are mixed together, the temperature of mixture is 12°C. When B and C are mixed, temperature of mixture is 21°C. Calculate temperature when A and C are mixed together

(1) 12.5°C

(2) 16°C

(3) 17.5°C

(4) 20°C

44) Two wires are fixed in a sonometer. Their tensions are in the ratio 8:1. The lengths are in the ratio 36:35. The diameters are in the ratio 4:1. Densities of the materials are in the ratio 1:2. If the higher frequency in the setting is 360 Hz, the beat frequency when the two wires sounded together is:

(1) 8

(2) 5

(3) 10

(4) 6

45) The wave function for a certain standing wave on a string fixed at both ends is $y(x,t) = 4 \sin(15\pi x) \cos 96\pi t$ where x and y are in meters and t is in seconds. The shortest possible length of the string is :-

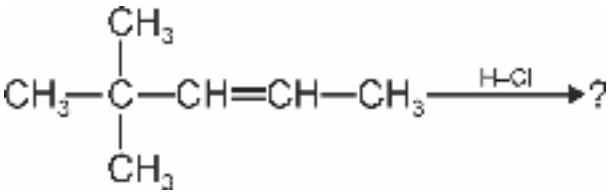
(1) 2.3 cm

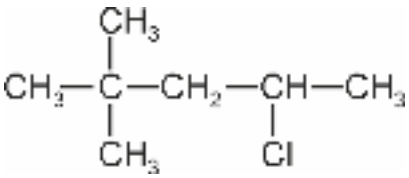
(2) 6.66 cm

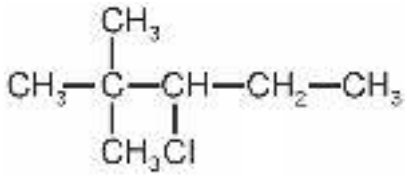
(3) 7.2 cm

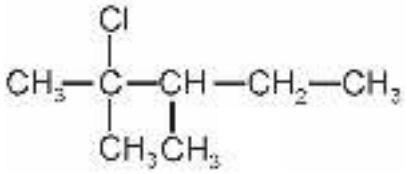
(4) 10.5 cm

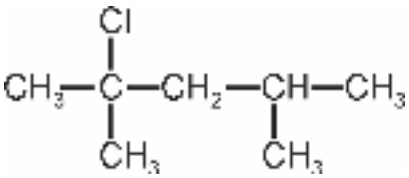
CHEMISTRY

1) 

(1) 

(2) 

(3) 

(4) 

2) Debromination of d-2, 3-dibromobutane gives:

(1) Trans-2-butene
(2) Cis-2-butene
(3) 1-butene
(4) 2-butyne

3) Which of the following compound has sp hybridised carbon -

(1) $CH_3 – CH = CH – CHO$
(2) $CH_3 – CO – CH_3$
(3) $CH_3 – (C)_2 – CH_3$
(4) $CHO \quad \ddot{N}H_2$

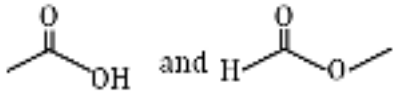
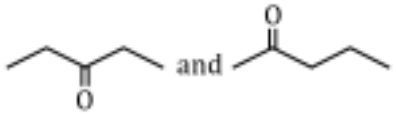
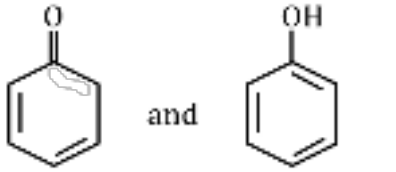
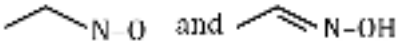
4) Total no of 'C' atoms in smallest alkane which contains only 1° and 3° 'C' atoms

(1) 2
(2) 3
(3) 4
(4) 5

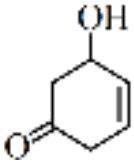
5) Paper chromatography is an example of

(1) Column chromatography
(2) Adsorption chromatography
(3) Partition chromatography
(4) Thin layer chromatography

6) Match the following :

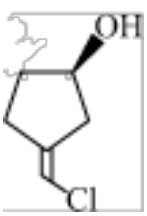
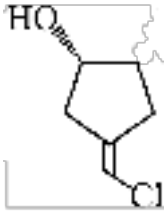
	Column I		Column II
a.		p.	Positional Isomers
b.		q.	Tautomer's
c.		r.	Functional Isomers
d.		s.	Metamers

(1) a → p ; b → r ; c → s ; d → q
(2) a → r ; b → p, s ; c → r, q ; d → r, q
(3) a → s ; b → q ; c → r ; d → p
(4) a → q ; b → p ; c → s ; d → r

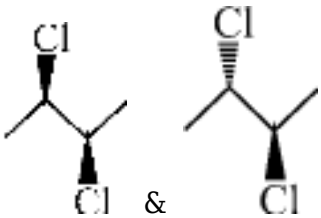
7) The correct IUPAC name of the given compound is 

(1) 7− hydroxy cyclohex −5− en −1− one
(2) 3− hydroxy cyclohex −5− en −1− one
(3) 8− hydroxy cyclohex −3− en −1− one
(4) 5− hydroxy cyclohex −3− en −1− one

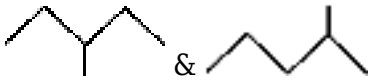
8) Pair of compounds with correct relation is :

(1)  & 
Geometrical isomers

(2) & Enantiomers

(3) 

Identical

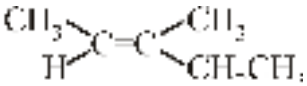
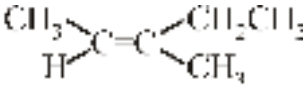
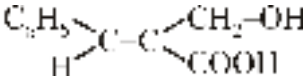
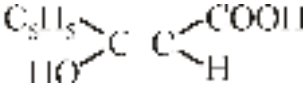
(4) 

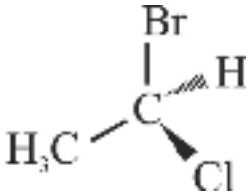
Chain isomers

9) According to CIP rule, the correct arrangement in order of decreasing priority is:-

(1) - OH > - CH_2OH > - CHO > - COOH
(2) - OH > - COOH > - CHO > - CH_2OH
(3) - COOH > - OH > - CHO > - CH_2OH
(4) - COOH > - CHO > - CH_2OH > - OH

10) The Z-isomer among the following is :-

(1) 
(2) 
(3) 
(4) 

11) Configuration of compound is :- 

(1) E
(2) R
(3) S
(4) Z

12) Reactivity order of HX in electrophilic addition reaction will be -

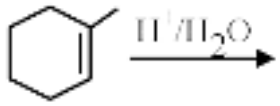
(1) HF < HCl < HBr < HI
(2) HF > HCl > HBr > HI
(3) HF < HCl < HI < HBr
(4) HCl < HF < HBr < HI

13) **Statement-I:** Oxymercuration-demercuration on alkene involves formation of intermediate

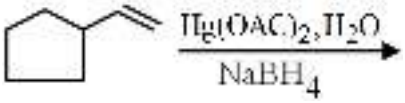
carbocation.
Statement-II: Oxymercuration-demercuration on alkene involves formation of alcohol.

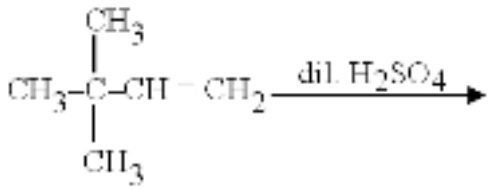
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

14) In which of the following reaction 2° alcohol will form

(1) 

(2) $CH_3-CH-CH=CH_2 \xrightarrow[H_2O_2/OH^{\ominus}]{B_2H_6}$

(3) 

(4) 

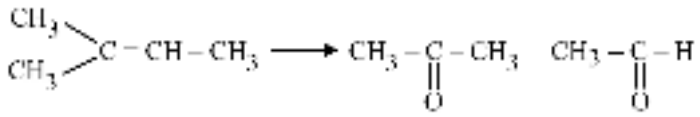
15) Ethene & Ethyne can be distinguished by

(1) Br_2/H_2O
(2) Baeyer's reagent
(3) Tollen's reagent
(4) All of above

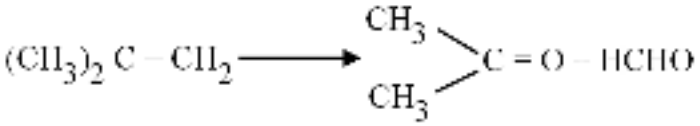
16) Which product will be formed on reductive ozonolysis of given compound. $CH_3 - CH = C = CH_2$

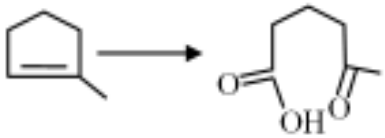
(1) Only Carbondioxide
(2) Only formaldehyde
(3) Only acetaldehyde
(4) Mixture of carbon dioxide, formaldehyde & acetaldehyde

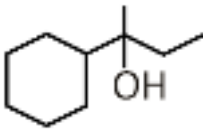
17) In which of the following reaction product is correct according to oxidative Ozonolysis

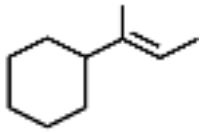
(1) 

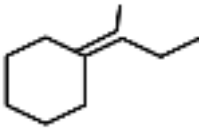
(2) $CH_3 - CH = CH_2 \rightarrow CH_3CHO + HCHO$

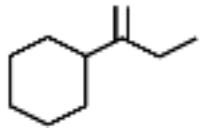
(3) 

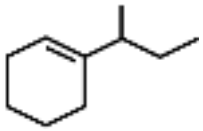
(4) 

18) Which of the following is not the product of dehydration of 

(1) 

(2) 

(3) 

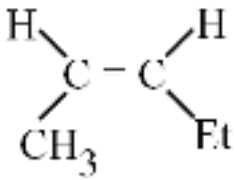
(4) 

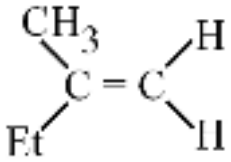
19) $CH_3 - C \equiv CH \xrightarrow[AgNO_3]{NH_4OH}$ Product (colour). The colour of product obtained is :-

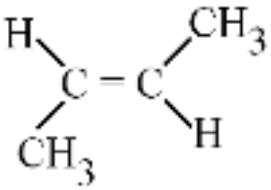
(1) White
(2) Red
(3) Yellow
(4) Colourless

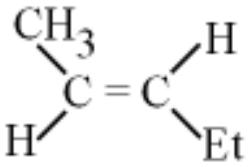
20) The product P is :-

$CH_3 - C \equiv C - CH_2 - CH_3 \xrightarrow{Na/liq.NH_3} P$

(1) 

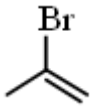
(2) 

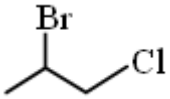
(3) 

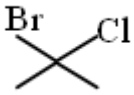
(4) 

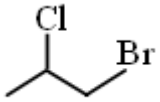
21) Product R is :-

$CH_3 - C \equiv CH \xrightarrow[1mol]{HBr} P \xrightarrow[1mol]{HCl} R$

(1) 

(2) 

(3) 

(4) 

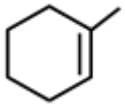
22) Number of σ and π bonds in product P is :-

$Et - X \xrightarrow{Na^{+}\bar{C}\equiv CH} \text{Product (P)}$

(1) 9σ, 2π
(2) 3σ, 2π
(3) 5σ, 1π
(4) 1σ, 1π

23) Most reactive towards free radical addition reaction:

(1)

(2) 

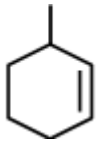
(3) $H_2C = CH_2$

(4)

24) Most reactive towards free radical substitution reaction?

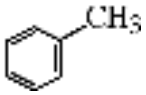
(1)

(2)

(3) 

(4) CH_4

25) Match the compound given in column-I with suitable reagent in column-II

	Column-I Compound		**Column-II Suitable reagent**
(A)	$H_2C=CH_2$	(p)	NBS
(B)		(q)	X_2/hv
(C)	CH_3–CH_2–CH_3	(r)	HBr+ H_2O_2

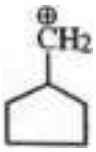
(1) (A)-(p); (B)-(q); (C)-(r)

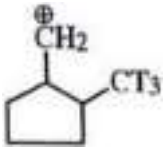
(2) (A)-(r); (B)-(p); (C)-(q)

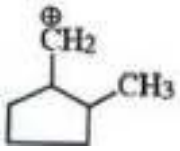
(3) (A)-(r); (B)-(p,q); (C)-(q)

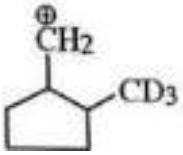
(4) (A)-(r); (B)-(q); (C)-(p)

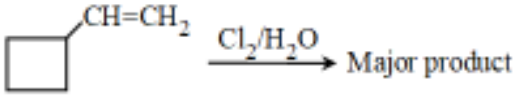
26) Which of the following carbocation is most stable ?

(1) 

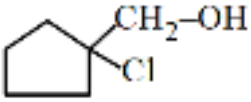
(2) 

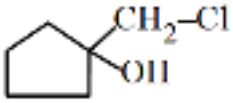
(3) 

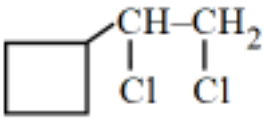
(4) 

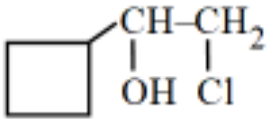
27) 

Find major product.

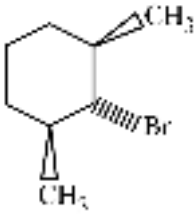
(1) 

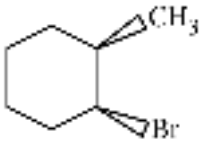
(2) 

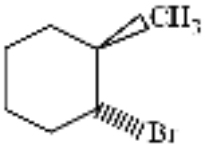
(3) 

(4) 

28) Ease of β-dehydrobromination among these substrates under the treatment of strong base.

(I) 

(II) 

(III) 

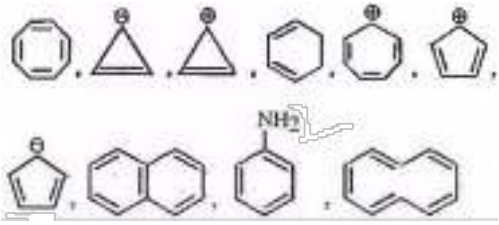
(1) I > II > III

(2) III > II > II

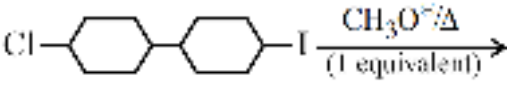
(3) II > I > III

(4) II > III > I

29) Amongst the following, the number of aromatic compound(s) are :



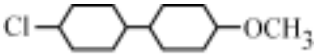
(1) 5
(2) 6
(3) 4
(4) 3

30) 

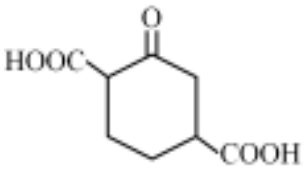
major product

(1)

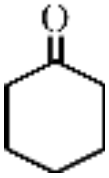
(2)

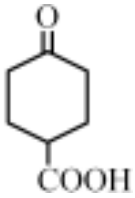
(3) 

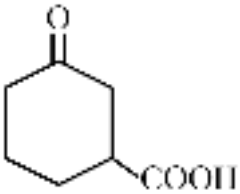
(4)

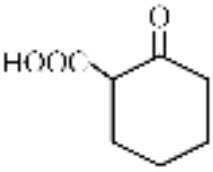
31) 

$\xrightarrow{\Delta}$ Product :-

(1) 

(2) 

(3) 

(4) 

32) Which of the following series contains electrophiles only ?

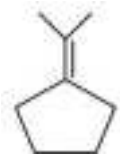
(1) H_2O, SO_2, $H_3O^{\oplus}$

(2) NH_3, H_2O, $AlCl_3$

(3) $AlCl_3, SO_3\ \overset{\oplus}{N}O_2$

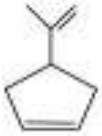
(4) $H_2O, \overset{\ominus}{Cl}, NH_3$

33) Which has minimum heat of hydrogenation ?

(1) 

(2)

(3)

(4) 

34) $Be_2C \xrightarrow{H_2O} CH_4 \xrightarrow[\text{(iii) } Cl_2/h\nu \text{ (iv) Red P/HI}]{\text{(i) } Cl_2/h\nu \text{ (ii) Na/Dry ether}} A.$

A is ?

(1) Ethane

(2) Propane

(3) Butane

(4) Isobutane

35) Which of the following reaction is called Wolf-kishner reduction ?

(1) $CH_3-\overset{O}{\overset{\|}{C}}-R \xrightarrow{NaBH_4} CH_3-\overset{OH}{\overset{|}{C}H}-R$

(2) $CH_3-\overset{O}{\overset{\|}{C}}-R \xrightarrow[\Delta]{N_2H_4/KOH} CH_3-CH_2-R$

(3) $CH_3-\overset{O}{\overset{\|}{C}}-R \xrightarrow[\Delta]{Zn\text{-}Hg/HCl} CH_3-CH_2-R$

(4) $CH_3-\overset{\overset{O}{\|}}{C}-R \xrightarrow{Ni/H_2} CH_3-\overset{\overset{OH}{|}}{CH}-R$

36) A and B related as :

$\alpha-\text{Butylene(A)} \xrightarrow{H_3O^+} X \xrightarrow{H^+/\Delta} B$

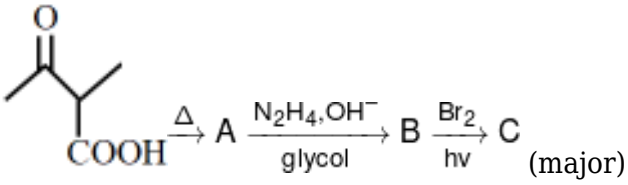
(1) Identical

(2) Chain isomer

(3) Position isomer

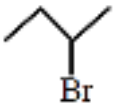
(4) Homologous

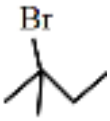
37) Predict the final product :

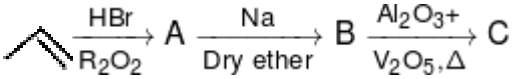


(1)

(2)

(3) 

(4) 

38) 

Correct for C.

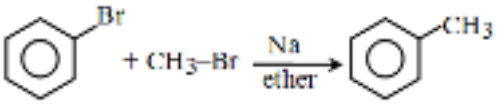
(1) It is aromatic

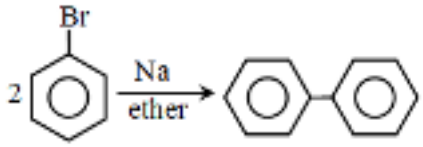
(2) It is benzene

(3) It is cyclic

(4) All of these

39)

	Column-I (Reaction)		**Column-II (Name of reaction)**
(A)		(P)	Wurtz reaction

(B)	CH_3–I+$(CH_3CH_2)_2CuLi$ → $CH_3CH_2CH_3$	(Q)	Fittig reaction
(C)		(R)	Corey-House reaction
(D)	$2CH_3–Br \xrightarrow[\text{ether}]{\text{Na}} CH_3–CH_3$	(S)	Wurtz-Fittig reaction

	A	B	C	D
(1)	P	Q	R	S
(2)	Q	P	R	S
(3)	Q	R	P	S
(4)	S	R	Q	P

(1) 1

(2) 2

(3) 3

(4) 4

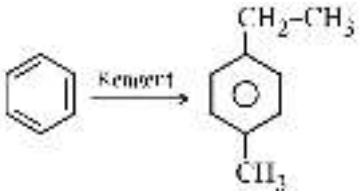
40) Arrange the following in their boiling points.
(i) *n*-butane
(ii) iso-butane
(iii) *n*-pentane
(iv) iso-pentane
(v) neopentane

(1) iii > i > ii > iv > v

(2) v > iv > ii > i > iii

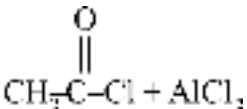
(3) iii > iv > v > i > ii

(4) ii > i > v > iv > iii



41) Give the correct order of reagent in following reaction respectively :-

(1) , CH_3 – Cl + $AlCl_3$, Zn – Hg / HCl

(2) Zn – Hg / HCl, CH_3Cl + $AlCl_3$, 

(3) $CH_3-\overset{O}{\overset{\|}{C}}-Cl + AlCl_3$, Zn-Hg/HCl,

$CH_3 - Cl + AlCl_3$

(4) All are correct

42) Friedel – Craft alkylation proceeds through the formation of electrophile -

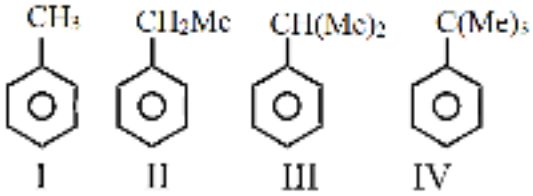
(1) Alkyl anion

(2) Alkyl free radical

(3) Alkyl cation

(4) Acylium cation

43) Correct order of reactivity of following for ESR (Electrophilic substitution reaction) is :-

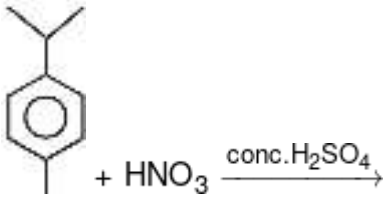


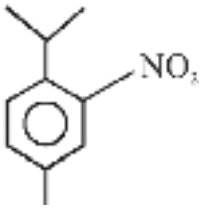
(1) IV > III > II > I

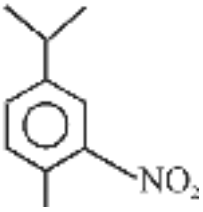
(2) IV > II > III > I

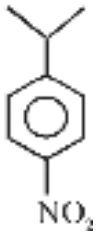
(3) I > II > III > IV

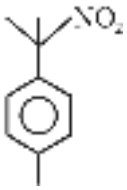
(4) I > IV > II > III

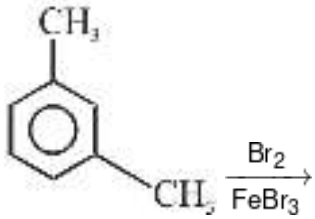
44) The major product formed in the reaction is : 

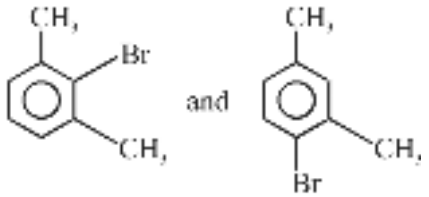
(1) 

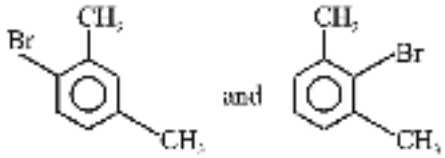
(2) 

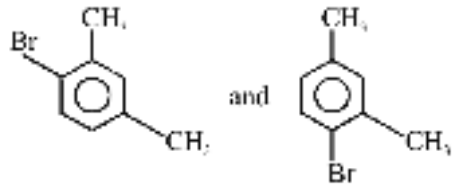
(3) 

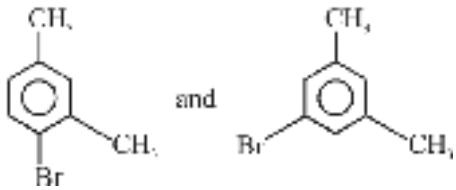
(4) 

45) What major products are formed when the following compound is treated with Br_2 in the presence of $FeBr_3$? 

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) (A) During the complete oxidation of pyruvate by the step wise removal of all the hydrogen atoms leaving three molecules of CO_2.
(B) In fermentation there is a net gain of two molecules of ATP and $NADPH_2$ from one molecule of glucose.
(C) The respiratiory quotient depends upon the type of respiratory substrate used during respiration.
(D) Pure proteins or fats are used as respiratory substrate.
Choose the option which has all wrong statements :-

(1) A, B, C

(2) A, C

(3) B, D

(4) B, C

2) Which one of the following statement is **incorrect** ?

(1) In fermentation less than 7% of the energy is released from one molecule of glucose.

(2) In alcohol fermentation only pyruvic acid decarboxylase enzyme is utilized.

(3) In fermentation the reducing agent is $NADH + H^+$ which is reoxidised to NAD^+.

(4) Yeasts poison themselves to death when the concentration of alcohol reaches about 13% in alcohol fermentation

3) Which of the following statement is **incorrect** ?

(1) During aerobic respiration, role of oxygen is limited to the terminal stage.

(2) In ETC (Electron Transport Chain), one molecule of $NADH + H^+$ gives rise to 3 ATP molecule and one $FADH_2$ gives rise to 2 ATP molecules.

(3) ATP is synthesized through complex IV.

(4) Glucose is favoured substrate for respiration

4) Which of the following statements are **correct** among the options given below ?
(i) Growth and differentation in plants is open.
(ii) Plant growth regulators are intracellular intrinsic factors affecting plant development.
(iii) Heterophylly in cotton, is an example of plasticity.
(iv) Sigmoid growth curve, is a characteristic of living organism growing in a natural environment.

(1) i, ii, iii

(2) ii, iii, iv

(3) i, iii, iv

(4) ii, iii

5) How many of the following statement/s is/are correct ?
(1) One single maize root apical meristem give rise to more than 17,500 new cells per minute.
(2) Watermelon may increase its cell size upto 17,500 times.
(3) Increase in length denotes the growth in pollen tube.
(4) An increase in surface area denotes the growth in a dorsiventral leaf.

(1) One

(2) Two

(3) Three

(4) Four

6) Which of the following statement is **incorrect** ?

(1) Ethylene promotes senescence and abscission of plant organs especially of leaves and flowers.

(2) Ethylene breaks seed and bud dormancy, initiates germination in peanut seeds.

(3) Ethylene promotes male flowers in cucumbers there by in increasing the yield.

(4) Ethylene is highly effective in fruit ripening.

7) Select the **incorrect** statement :-

(1) C_3 plants responds to higher CO_2 concentration by showing increased rate of photosynthesis.

(2) Tropical plants have a higher temperature optima than temperate plants for photosynthesis.

(3) C_4 plants show much lower temperature optima than C_3 plants for photosynthesis.

(4) At high light intensities both C_3 and C_4 plants show increase in rate of photosynthesis.

8) Consider the following four statements A,B,C and D. Select the right option for two correct statements :-
(A) Z scheme of light reaction takes place in presence of PS I only
(B) Only PS I is functional in cyclic photophosphorylation
(C) Cyclic photophosphorylation results into synthesis of ATP and $NADPH_2$
(D) Stroma lamellae lack PS II as well as NADP reductase enzyme

(1) C and D

(2) B and D

(3) A and B

(4) B and C

9) Identify the correct statements regarding Calvin cycle :-
(a) Regeneration is most crucial step.
(b) Carboxylation is catalysed by the enzyme RuBP carboxylase.
(c) Regeneration steps require 1 ATP for phosphorylation to form one RUBP.
(d) First stable product of CO_2 fixation is a C_3 acid.
(e) The primary acceptor is a 2 - carbon compound.
Choose the correct answer from the options given below :

(1) (a), (b) and (e)

(2) (c), (d) and (e)

(3) (b), (c) and (d)

(4) (b), (c) and (e)

10)

Identify the correct match from columns I, II and III :-

	Column-I		**Column-II**		**Column-III**
1	Chlorophyll 'a'	a	Absorption peak at 680 nm	i	Prevents Photo-oxidation
2	Photosystem-I	b	Accessory pigments	ii	Only grana Lamellae
3	Photosystem-II	c	Chief pigment	iii	Blue green in colour
4	Carotenoids	d	Absorption peak at 700 nm	iv	Present in both grana and stroma lamellae

(1) 1 - c - iii, 2 - d - iv, 3 - a - ii, 4 - b - i

(2) 1 - b - i, 2 - a - ii, 3 - c - iii, 4 - d - iv

(3) 1 - b - iv, 2 - c - iii, 3 - d - ii, 4 - a - i

(4) 1 - b - i, 2 - a - ii, 3 - d - iii, 4 - c - iv

11) Match the column :

Column-I		Column-II	
(A)	Joseph Priestley	(i)	Green plant parts synthesise sugar
(B)	Jan Ingenhousz	(ii)	In photosynthesis O_2 comes from H_2O
(C)	Julius von Sachs	(iii)	Discovered oxygen
(D)	Cornelius Van Niel	(iv)	Sunlight is essential to the plant process that purifies fouled air

(1) A = (i), B = (iv), C = (iii), D = (ii)

(2) A = (iv), B = (ii), C = (iii), D = (i)

(3) A = (ii), B = (i), C = (iii), D = (iv)

(4) A = (iii), B = (iv), C = (i), D = (ii)

12) Match column-I with column-II.

Column-I		Column-II	
(I)	IAA	(A)	Terpenes
(II)	GA	(B)	Indole compounds
(III)	ABA	(C)	Adenine derivatives
(IV)	Kinetin	(D)	Carotenoid derivatives

The **correct match** is :-

(1) I - B, II - D, III - C, IV - A

(2) I - B, II - A, III - D, IV - C

(3) I - C, II - A, III - D, IV - B

(4) I - D, II - B, III - A, IV - C

13) Match Column-I with Column-II and select the correct option from the codes given below :-

	Column-I		Column-II
A.	Auxins	(i)	Breaking seed dormancy
B.	Gibberellins	(ii)	Inducing fruit ripening
C	Cytokinins	(iii)	Formation of abscission layer
D.	Ethylene	(iv)	Root initiation
		(v)	Chloroplast development in leaves

(1) A-(iv), B-(i), C-(v), D-(ii)

(2) A-(iv), B-(v), C-(iii), D-(ii)

(3) A-(i), B-(iii), C-(ii), D-(iv)

(4) A-(iii), B-(iv), C-(i), D-(v)

14) Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion (A) : There are two phosphorylations in a glycolysis process with the help of ATP.
Reason (R) : In glycolysis glucose converts into glucose-6-phosphate and fructose-6-phosphate converts into fructose 1,6 bisphosphate by the use of ATP.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

15) **Assertion (A) :-** It is better to consider respiratory pathway as an amphibolic pathway.
Reason (R) :- Respiratory pathway comes into the picture both during breakdown and synthesis of substrates.

(1) Assertion is true but reason is false
(2) Both (A) and (R) are true but (R) is not correct explanation of (A)
(3) Both (A) and (R) are false
(4) Both (A) and (R) are true and (R) is correct explanation of (A)

16) **Assertion (A) :-** The respiratory quotient (RQ) of carbohydrates is generally 1.
Reason (R) :- During carbohydrate respiration, the amount of CO_2 released is equal to the amount of O_2 consumed.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

17) **Statement-I:** The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2 takes place in cytoplasm.
Statement-II: The passing on of the electron removed as part of the hydrogen atom to molecular O_2 with simultaneous synthesis of ATP take place in the inner membrane of mitochondria.

(1) Statement I and II both are correct.
(2) Statement I and II both are incorrect.
(3) Only Statement I is correct.
(4) Only Statement II is correct.

18) **Statement-I :** The carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in cell.
Statement-II : Plants, unlike animals, have no specialised organs for gaseous exchange but they have stomata and lenticels for this purpose.

(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statement I and II are correct

(4) Both statement I and II are incorrect.

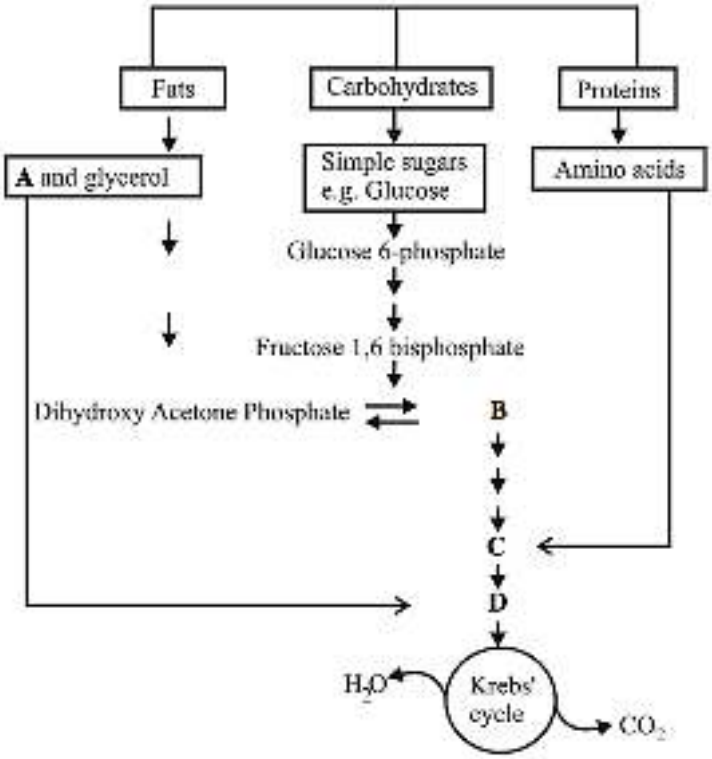
19) Total number of ATP molecules formed during complete oxidation of fructose 1, 6-bisphosphate is :

(1) 20
(2) 32
(3) 36
(4) 40

20) How many oxygen atoms will be required in ETS to oxidise intermediates generated only in mitochondrial matrix during breakdown of a sucrose molecule ?

(1) 5
(2) 10
(3) 15
(4) 20

21) Given below is the diagrammatic sketch of respiratory pathways. Identify the compounds labelled A, B, C and D and select the right option about them:



	A	B	C	D
(1)	Pyruvic acid	Acetyl CoA	Fatty acid	Glyceraldehyde –3–phosphate
(2)	Glyceraldehyde –3–phosphate	Fatty acid	Acetyl CoA	Pyruvic acid

(3)	Fatty acid	Glyceraldehyde –3–phosphate	Pyruvic acid	Acetyl CoA
(4)	Fatty acid	Pyruvic acid	Glyceraldehyde –3–phosphate	Fatty acid

(1) 1
(2) 2
(3) 3
(4) 4

22) The number of substrate level phosphorylations in two turns of krebs cycle is:

(1) 3
(2) Zero
(3) 1
(4) 2

23) To produce one fructose, respectively how many ATP, $NADPH_2$ and CO_2 are required during dark reaction of C_4 plants ?

(1) 36, 18 and 12
(2) 18, 12 and 6
(3) 30, 12 and 6
(4) 30, 18 and 12

24) The following (P to U) are the main steps of chemiosmotic ATP synthesis in the light reaction. Which option places them in correct order ?
(P) H^+ concentration gradient established
(Q) H^+ diffuses through ATP synthase
(R) Carriers use energy to move H^+ across the membrane
(S) Electrons from PSII pass along electron transport chain
(T) Light excites electrons in PSII
(U) Energy of H^+ flow is used by ATP synthase to make ATP

(1) PQTSRU
(2) STPQRU
(3) TSRPQU
(4) TSRUQP

25) According to 'absorption spectrum' graphical study, the rate of absorption of blue light by the following pigments, in ascending order, is as follows :
(A) Chlorophyll - a
(B) Chlorophyll - b
(C) Carotenoids

(1) A→B→C
(2) B→A→C

(3) C→A→B

(4) C→B→A

26) In calvin cycle, for synthesis of three molecules of sucrose, the requirement of ATP and NADPH is respectively.

(1) 108 and 72

(2) 54 and 36

(3) 30 and 12

(4) 60 and 24

27) Read the following statements.
Statement 1: In membrane system of chloroplast , enzymatic reactions synthesise sugar, which in turn forms starch.
Statement 2: The stroma is responsible for trapping the light energy and also for the synthesis of ATP and NADPH.

(1) Both statements are correct

(2) only statement 1 is correct

(3) only statement 2 is correct

(4) Both statements are incorrect

28) **Assertion :** There is a not a complete one- to-one overlap between the absorption spectrum of chlorophyll a and the action spectrum of photosynthesis
Reason : Most of the photosynthesis takes place in the blue and red regions of the spectrum but some photosynthesis does take place at the other wavelengths of the visible spectrum.

(1) Assertion and reason both are true but reason is not the correct explanation of the assertion

(2) only Assertion is true

(3) Both assertion and reason are false

(4) Assertion and reason both are true and reason is the correct explanation of the assertion

29) Which of the following pigment shows yellow green colour on paper chromatography ?

(1) Chlorophyll-a

(2) Chlorophyll-b

(3) Xanthophyll

(4) Carotenoids

30) Half leaf experiment proves that :-

(1) Light is essential for photosynthesis.

(2) CO_2 is essential for photosynthesis.

(3) O_2 release during photosynthesis.

(4) Chlorophyll is essential for photosynthesis.

31) Who found that in green part of plant, glucose is made and stored as starch ?

(1) Sachs
(2) Ingenhousz
(3) Niel
(4) Engelmann

32) In light reaction, plastoquinone facilitates the transfer of electron from :-

(1) PS-I to ATP synthase
(2) PS-II to cyt b_6f complex
(3) Cyt b_6f complex to PS-I
(4) PS-I to $NADP^+$

33)

Match the column-I and column-II and select the correct option :-

Column-I		**Column-II**	
A	ATP	I	Not an electron carrier but a H carrier
B	PQ	II	Diffeses out of the chloroplast
C	O_2	III	Formed in cyclic and non cyclic photophosphorylation
D	$NADP^+$	IV	Terminal acceptor of electron is non cyclic photophosphorylation

(1) A-IV , B-I , C-II , D-III
(2) A-III , B-II , C-I , D-IV
(3) A-II , B-I , C-III , D-IV
(4) A-III , B-I , C-II , D-IV

34)

Which of the following is not a C_4-plant ?

(1) Maize
(2) Sugarcane
(3) Sorghum
(4) Rice

35) **Statement-I:** Plant follows different pathways in response to environment or phases of life to form different kinds of structures.
Statement-II: In cotton, leaves of juvenile phase plant are different in shape from those in mature plant.

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

36) Which of the following is synthesised by dedifferentiation ?

(1) Epidermis
(2) Cortex
(3) Cork cambium
(4) Pith

37) Read the following roles of plant hormones carefully.
(A) It acts as an antagonist to GAs.
(B) It acts as a general plant growth inhibitor and an inhibitor of plant metabolism.
(C) It promotes seed germination.
(D) It stimulates the closure of stomata and increases the tolerance of plants to various kinds of stresses.
How many above statements are related with ABA ?

(1) Four
(2) Three
(3) Two
(4) One

38) Match the columns & select the correct option -

	Column-I		**Column-II**
a.	C.Darwin and F. Darwin	i.	Cytokinin
b.	Miller and Skoog	ii.	Gibberellins
c.	Cousins	iii.	Ethylene
d.	Kurosawa	iv.	Auxin

(1) a-ii, b-iii, c-i, d-iv
(2) a-iv, b-i, c-iii, d-ii
(3) a-iii, b-ii, c-iv, d-i
(4) a-i, b-ii, c-iii, d-iv

39)

Cells positioned away from ___(A)___ apical meristem differentiate as root-cap cells, while those pushed to the periphery mature as ___(B)___ .

(1) (A) shoot (B) Epidermis
(2) (A) Root (B) Epidermis
(3) (A) shoot (B) Endodermis
(4) (A) Root (B) Endodermis

40) Which of the following is not correctly matched ?

(1) Auxin-controls xylem differentiation
(2) Ethephon-thinning of cotton and cherry
(3) Cytokinin-Hastens the maturing period in conifers.
(4) Gibberellins-delay senescence

41) Which acts as a mobile carrier for the transfer of electrons between complex-II and complex-III ?

(1) Cytochrome-C
(2) Ubi quinone
(3) Plastocyanin
(4) FAD

42) Which of the following is released is a by product in the link reaction ?

(1) Oxygen
(2) ATP
(3) Water
(4) CO_2

43) Which of the following is first member of krebs cycle/CA cycle ?

(1) Citric acid
(2) Acetyl Co-A
(3) Pyruvic acid
(4) Oxalo acetic acid (O.A.A.)

44) Read the following statement on plant growth & development.
(A) Oxygen helps in releasing metabolic energy essential for growth activities.
(B) Environmental signals such as light and gravity also affect certain phases of growth.
(C) Parthenocarpy can be induced by auxins.
(D) ABA is a plant growth promoter.
(E) Water provides the medium for enzymatic activities needed for growth.
Choose the option with all correct statement.

(1) A, B, C only
(2) B,C,D only
(3) A, B,C,E only
(4) All of these

45) **Statement-I :** Auxin was isolated by Charles Darwin and his son Francis Darwin when they observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism)
Statement-II : The bakane (foolish seedling) disease of rice seedling was caused by a fungal pathogen *Gibberella Fujikuroi*. E. Kurosawa reported the appearance of symptoms of the disease in rice seedlings when they were treated with sterile filtrates of the fungus.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

46) Select the incorrect statement :

(1) Cerebral cortex, greyish in appearance

(2) Concentrated neuron cell body gives grey color to the cerebral cortex

(3) Fibres of the tract which are covered with myelin sheath, constitute inner part of cerebral hemisphere

(4) Cerebrum is wrapped around the structure called medulla

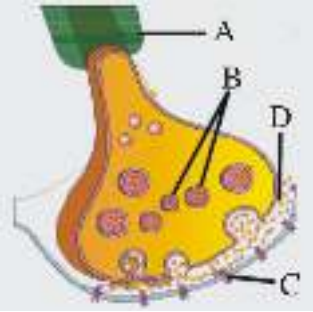
47) The somatic neural system release impulse from the ____A____ to ___B___.
Fill in the blanks (A & B) with appropriate words.

(1) A-CNS; B-smooth muscles

(2) A-PNS; B-skeletal muscles

(3) A-CNS; B-skeletal muscles

(4) A-CNS; B-Visceral organs

48) Select the correct option: 

(1) A-Synaptic vesicles

(2) B-Post-synaptic membrane

(3) C-Axon

(4) D-Synaptic cleft.

49) Given below are two statements
Statement-I : Axonal membrane is comparatively more permeable to potassium ions (K^+) and nearly impermeable to sodium ions (Na^+)
Statement-II : Outer surface of the axonal membrane possess a positive charge while its inner surface negative charged known as polarised state.

(1) Statement-I and II are correct

(2) Statement-I is correct but II is incorrect

(3) Statement-I is incorrect but II is correct

(4) Statement-I and II are incorrect

50) Which of the following structures or regions is incorrectly paired with its function -

(1) Association area : Functions like intersensory associations, memory & communication.

(2) Hypathalamus : Consists of fibre tracts that interconnect different regions of brain.

(3) Corpus callosum : Band of fibres connecting Left and Right Cerebral Hemispheres.

(4) Medulla oblongata : Controls respiration and cardiovascular reflexes.

51) Match the components of the brain with their respective descriptions :-

(i)	Pons	(A)	Has a convoluted surface to accommodate more neurons
(ii)	Cerebellum	(B)	Integrates auditory and visual signals
(iii)	Medulla oblongata	(C)	Contains fibre tracts that interconnect different brain regions
(iv)	Midbrain	(D)	Connected to the spinal cord

(1) i-A, ii-B, iii-C, iv-D

(2) i-A, ii-D, iii-C, iv-B

(3) i-C, ii-A, iii-D, iv-B

(4) i-D, ii-C, iii-B, iv-A

52) How many statements are not incorrect :-
(a) Brain controls the voluntary movements
(b) Brain controls the functioning of involuntary organs.
(c) Brain controls the thermoregulation, hunger and thirst
(d) Brain controls the activities of several endocrine glands

(1) a only

(2) b only

(3) a and d only

(4) a, b, c and d

53) Match column-I with column-II

Column-I		**Column-II**	
(i)	Neurons	(a)	Network of neuron simply
(ii)	Hydra	(b)	Highly specialised cells
(iii)	Arthropoda	(c)	Brain and more neuron develops
(iv)	Vertebrates	(d)	Brain made of ganglia and neural tissue

(1) (i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)

(2) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)

(3) (i)-(b), (ii)-(a), (iii)-(d), (iv)-(c)

(4) (i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)

54) **Assertion (A) :** The depolarization phase of an action potential is primarily driven by sodium influx.

Reason (R) : Sodium voltage channels open in response to a threshold stimulus, allowing sodium ions to rush into the neuron.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

(3) (A) is correct and (R) is incorrect.

(4) (A) is incorrect and (R) is correct.

55) Which of the following statement is not correct.
(A) Electrical potential difference across the polarized plasma-membrane is known as action potential in the resting stage.
(B) Cl^- ion is responsible for generating an action potential.
(C) The impulse is called action potential.

(1) A and B

(2) B and C

(3) A and C

(4) A, B and C

56) Consider the following four statement (A-D) and select the option which includes the correct one only.
(A) The gap between two myelinated sheath is called nodes of ranivir.
(B) Myelinated nerve fiber are present in spinal and cranial nerve only.
(C) Unmyelinated nerve fiber is enclosed by a schwann cell that does not form a myelin sheath around the axon. It is commonly formed in autonomous and somatic neural system.
(D) Each branch of axon terminates put a bulb like structure called synaptic knob which possess synaptic vessicle that contains chemical (Neurotransmeter).

(1) A, C, D

(2) A, B

(3) B, D

(4) B, C, D

57) Read column-I, II and III and identify their correct match -

Column-I		Column-II		Column-III	
(A)	Fore brain	(P)	Corpora quadrigemina	(i)	Controls respiration and cardiovascular reflex
(B)	Mid brain	(Q)	Corpus callosum	(ii)	Regulation of temperature hunger and thrust
(C)	Hind brain	(R)	Hypothalamus	(iii)	Connecting left & right cerebral hemisphere
(D)	Tract of nerve fibres	(S)	Medulla oblongata	(iv)	Four round swellings (Lobes)

(1) (A)-(Q)-(ii), (B)-(P)-(iv), (C)-(R)-(i), (D)-(S)-(iii)

(2) (A)-(R)-(ii), (B)-(P)-(iv), (C)-(S)-(i), (D)-(Q)-(iii)

(3) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(4) (A)-(R)-(iv), (B)-(P)-(iii), (C)-(S)-(ii), (D)-(Q)-(i)

58) Read the following terminology of column-I & column-II and identify their correct match -

Column-I		Column-II	
(A)	Piamater	(i)	Retina
(B)	Grey matter	(ii)	Pons
(C)	Neuro secretory cells	(iii)	Cerebral cortex
(D)	Bipolar neuron	(iv)	Cranial meninx
		(v)	Cerebral aqueduct
		(vi)	Hypothalamus
		(vii)	Corpus calloasum

(1) (A)-(iv), (B)-(iii), (C)-(vi), (D)-(i)

(2) (A)-(iii), (B)-(i), (C)-(vi), (D)-(vii)

(3) (A)-(iv), (B)-(ii), (C)-(v), (D)-(vi)

(4) (A)-(ii), (B)-(iv), (C)-(vi), (D)-(i)

59) Read the following statements -
(A) The cerebral cortex is referred is as grey matter due to its greyish appearance.
(B) The hindbrain comprises pons, thalamus and medulla.
(C) Midbrain forms the major part of human brain.
(D) Action potential is known as nerve impulse.
(E) Pons consists of centres which control cardiovascular reflex and gastric secretions.
Choose the option having all correct statements -

(1) B and C

(2) B, C and E

(3) A, D and E

(4) A and D

60) Match the Column-I with Column-II and choose correct option :

	Column-I (Gland)		Column-II (Function)
A.	Pineal	1.	Regulates biological clock
B.	Hypothalamus	2.	Controls pituitary secretion
C.	Posterior pituitary	3.	Release oxytocin and ADH
D.	Anterior pituitary	4.	Secretes GH, TSH, FSH etc.

(1) A-1, B-3, C-4, D-2

(2) A-3, B-1, C-4, D-2

(3) A-1, B-2, C-3, D-4

(4) A-4, B-3, C-3, D-4

61)

In which one of the following options Column I is correctly matched with Column II?

	Column I	Column II
(1)	Hormone	Nutritive chemical
(2)	Kidney	Non organized gland
(3)	Pituitary gland	Non organized gland
(4)	Invertebrates	Complex endocrine systems

(1) 1

(2) 2

(3) 3

(4) 4

62) Match List-I with List-II

List-I		List-II	
a	Steroid	I	Epinephrine
b	Amino acid derivative	II	intracellular receptor
c	Polypeptide	III	extra cellular receptor
d	FSH	IV	progesterone
e	Thyroid hormone	V	Insulin

(1) a-IV; b-I; c-V; d-III; e-II

(2) a-IV; b-I; c-IV; d-II; e-III

(3) a-I; b-V; c-IV; d-III; e-II

(4) a-IV; b-V; c-I; d-II; e-III

63) Most of hormone secreted pituitary gland are_____in nature whereas most of hormones of adrenal cortex are_____ in nature.

(1) Protein and protein respectively

(2) Protein and amino acids respectively

(3) Amino acid and glycoproteins respectively

(4) protein and steroids respectively

64) Match List-I with List-II.

List-I		List-II	
A	Diabetes insipidus	I	Melatonin
B	Grave's disease	II	impaired secretion of ADH

C	Diabetes mellitus	III	Hyperthyrodism
D	Disturbed diurnal rhythm	IV	Prolonged hyperglycema

(1) A-IV; B-III; C-II; D-I

(2) A-II; B-III; C-IV; D-I

(3) A-II; B-III; C-I; D-IV

(4) A-IV; B-III; C-I; D-II

65) Match the List-I with List-II

List-I		**List-II**	
A	Cholecystokinin	i	Juxtaglomerular cells of kidney
B	Erythropoietin	ii	Leydig cells
C	Progesterone	iii	Stimulates the secretion of pancreatic enzyme and bile juice
D	Testosterone	iv	Corpus luteum

Choose the correct answer from the given options below:

(1) A-i, B-iii, C-iv, D-ii

(2) A-i, B-ii, C-iii, D-iv

(3) A-iii, B-i, C-ii, D-iv

(4) A-iii, B-i, C-iv, D-ii

66) __A__ is a steroid hormone and __B__ is a peptide hormone :

(1) A-Cortisol, B-Estradiol

(2) A-Estradiol, B-Cortisol

(3) A-Cortisol, B-Insulin

(4) A-Insulin, B-Cortisol

67) Given below are two statements :
Assertion : Insulin is a hypoglycemic hormone.
Reason : Insulin enchances cellular glucose uptake and utilisation.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

68) Hormones are ___A___ chemicals which acts as ___B___ messengers and are produced in ___C___ amounts.

	A	**B**	**C**
(1)	Nutrient	Intracellular	Trace

(2)	Nutrient	Inter-cellular	Trace
(3)	Non-nutrient	Intercellular	Trace
(4)	Non-nutrient	Intercellular	large

(1) 1

(2) 2

(3) 3

(4) 4

69) Read the following statements carefully and choose correct option as answer :-
(A) Melatonin helps in maintaining normal body temperature.
(B) Melatonin influences metabolism.
(C) Melatonin secretion is suppressed by bright light.
(D) Melatonin regulates sleep wake cycle.

(1) Only (A) and (B) are incorrect.

(2) (A), (B), (C) and (D) are incorrect.

(3) All statements (A) to (D) are correct.

(4) (A), (C) are (D) incorrect.

70) Given below are two statements :
Statement-I : Adrenal medulla hormones increase alertness, pilo erection, sweating & heart beat.
Statement-II : Insulin resistance results in a disease called diabetes mellitus.

(1) Both statement I and statement II are incorrect

(2) Both statement I and statement II are correct

(3) Statement I is correct and statement II is incorrect

(4) Statement I is incorrect and statement II is correct

71) Match the List-I with List-II

List-I		**List-II**	
A	para thyroid hormone	I	blood calcium level
B	epinephrine and nor epinephrin	II	Adrenal medulla
C	Thyroid stimulating hormone	III	Anterior pituitary
D	melatonin	IV	Sleep wake cycle

Choose the correct answer from the options given below

(1) A-III, B-IV, C-I, D-II

(2) A-III, B-II, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-I, C-II, D-IV

72) With regard to hormones, identify the correct statements.
(A) Hormones that interact with intracellular receptors mostly not regulates gene expression
(B) Epinephrine is a aminoacid derivative hormone

(C) Hormone that interact with membrane bound receptors normally do not enter target cell, but not generate second messanger
(D) progesterone is a steroid hormone.
Choose the most appropriate answer from the options given below.

(1) C and E only
(2) C and D only
(3) B and C only
(4) B and D only

73) Thyroid gland also secretes a _A_ hormone called thyrocalcitonin which regulate the blood _B_ levels.

	A	B
(1)	Lipid	Calcium
(2)	Steroid	Sodium
(3)	Protein	Calcium
(4)	Amino acid	Calcium

(1) 1
(2) 2
(3) 3
(4) 4

74) **Statement-I:** Parathyroid hormone decrease the Ca^{+2} level in the blood and increase density of bone.
Statement-II: Parathyroid hormone along with thymosin, play a significant role in calcium balance in the body.
Read the following statement and selective right answer.

(1) Statement-I is correct but statement II is false
(2) Statement-I is false but statement II is correct
(3) Both statement I and statement II are true
(4) Both Statement I and statement II are false

75) Fill in the blanks :
__A__ regulate female sexual behaviour. __B__ supports pregnancy. __C__ acts on the mammary glands and stimulates the formation of alveoli and milk secretion.
Choose the correct option :

(1) A-Progesterone, B-Estrogen, C-Estrogen
(2) A-Estrogen, B-Progesterone, C-Progesterone
(3) A-Estrogen, B-Progesterone, C-Estrogen
(4) A-Progesterone, B-Progesterone, C-Estrogen

76) Given below are two statements
Assertion : Ovary is composed of ovarian follicles and testis in composed of seminiferous tubules.
Reason : Both ovary and testis are primary sex organ.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

77) The common features which are present in muscles & nerves.
(A) Excitability
(B) Conductivity
(C) Contractability
(D) Sensitivity

(1) A, B and D
(2) A, B and C
(3) B, C and D
(4) A, C and D

78) **Assertion :-** Red muscles are also called aerobic muscles.
Reason :- Red muscles also contain plenty of mitochondria which can utilised the large amount of oxygen stared in them for ATP production.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

79) The muscles located in the inner walls of the digestive tract are __A__ and __B__.

	A	**B**
(1)	Striated	Involuntary
(2)	Nonstriated	Voluntary
(3)	Nonstriated	Involuntary
(4)	Striated	Voluntary

(1) 1
(2) 2
(3) 3
(4) 4

80) Given below are two statements : one is labelled as Assertion 'A' and other as Reason 'R'.
Assertion (A) : Sarcoplasmic reticulum of the muscle fibre is the store house of calcium ions.
Reason (R) : Muscle fibre is a syncytium as the sarcoplasm contain many nucleus.

in the light of the above statements, choose the correct answer from the options given below :

(1) A and R both statement are correct, R is correct explanation of 'A'.

(2) A and R both statement are correct, R is not correct explanation of 'A'.

(3) A is correct but R is false.

(4) A is false but R is true.

81) Given below are two statements : one is labelled as Assertion 'A' and other as Reason 'R'.
Assertion (A) : Cells of the human body exhibit three main types of movements, namely amoeboidal, ciliary, muscular.
Reason (R) : Movement is one of the significant features of living beings.
in the light of the above statements, choose the correct answer from the options given below :

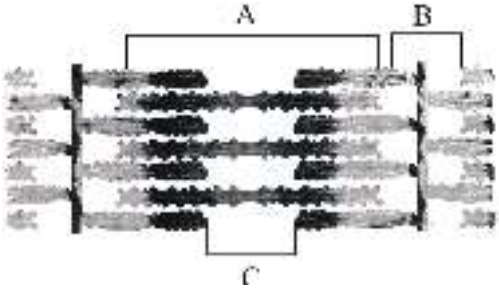
(1) Both A and R are correct R is correct explanation of A.

(2) A is false but R is true.

(3) A is true but R is false.

(4) Both A and R are true R is not correct explanation of A.

82) Choose the most appropriate option regarding length of labelled regions A, B and C during



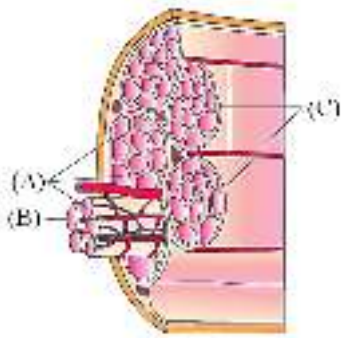
muscle contraction in the given structure of sarcomere :

(1) B becomes short while A and C remain same

(2) A and C become short while B remains same

(3) B and C become short while A remains same

(4) A becomes short while B and C remain same



83) Identify the figure and choose the correct matching :-

(1) A → muscle fibre, B → sarcolemma, C → fascicle

(2) A → muscle bundle , B → Sarcolemma, C → muscle cell

(3) A → Muscle fibre, B → Blood capillary, C → muscle bundle

(4) A → fascicle, B → sarcolemma, C → muscle fibre

84) Read the following statements carefully identify how many of them correct ?
(i) Cardiac muscle are the muscle of heart.
(ii) Cardiac muscle are involuntary in nature.
(iii) Cardiac muscle are striated, branched.
(iv) Cardiac muscle cell are assemble in a branching pattern to form a cardiac muscle.
(v) Cardiac muscle help in the transportation of food through the digestive tract.

(1) 4
(2) 1
(3) 2
(4) 5

85) Which of the following is wrong match

(1) Ball and socket joint – Between humerus and pectoral girdle
(2) Hinge joint – Knee joint
(3) Saddle joint – Carpometacarpal joint of thumb
(4) Pivot joint – Humerus with radius joint

86) Math the column

	Disorders		**Key words**
P	Myathenia gravis	a	Inflammation due to uric acid
Q	Muscular dystrophy	b	Increase chances of fractures
R	Osteoporosis	c	Genetic disorder
S	Gout	d	Autoimmune disorder

(1) P–b, Q–a, R–c, S–d
(2) P–a, Q–b, R–c, S–d
(3) P–d, Q–c, R–b, S–a
(4) P–d, Q–c, R–a, S–b

87) Choose the option which is **not correct** w.r.t. the disorders of muscular and skeletal system.

(1) Arthritis is the inflammation of joints
(2) Gout is the inflammation of joints due to accumulation of lactic acid crystals
(3) Muscular dystrophy is the progressive degeneration of skeletal muscle mostly due to genetic disorder
(4) Osteoporosis is an age-related disorder characterised by decreased bone mass and increased chances of fractures

88) **Statement I :** In mammals, the largest vertebra is lumbar.
Statement II : The number of lumbar vertebrae in human vertebral column is five.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

89) **Assertion (A) :** Osteoporosis mainly occurs in post menopausal women .
Reason (R) : Estrogen level decreases in postmenopausal women.

(1) A and R both are correct and R is the correct explanation of A.

(2) A and R both are correct and R is not the correct explanation of A.

(3) A is incorrect but R is correct.

(4) A is correct but R is incorrect.

90) Match the List-I and List-II

	List-I		**List-II**
(A)	Fibrous Joint	(i)	Considerable movement
(B)	Cartilaginous Joint	(ii)	Permits limited movement
(C)	Synovial joints	(iii)	Do not allow any movement.
(D)	Pivot Joint	(iv)	Between atlas and axis

Choose the correct answer from the option given below.

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-iii, C-ii, D-i

(3) A-iii, B-ii, C-i, D-iv

(4) A-iii, B-iv, C-i, D-ii

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	2	4	2	2	3	4	1	2	2	3	4	3	2	1	1	3	3	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	4	3	4	4	2	4	1	3	3	1	1	2	2	2	3	4	2	3
Q.	41	42	43	44	45															
A.	4	2	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	3	3	3	2	4	1	2	2	2	1	4	3	3	4	4	4	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	1	2	3	3	2	4	4	1	2	3	3	1	1	2	3	3	4	4	3
Q.	86	87	88	89	90															
A.	3	3	3	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	3	3	2	3	3	2	3	1	4	2	1	4	4	4	4	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	3	3	1	4	4	2	2	1	2	4	4	3	3	2	2	2	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	3	3	4	3	4	1	2	3	4	3	1	1	1	2	1	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	1	4	2	4	3	1	3	3	2	3	4	3	4	2	2	1	1	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	1	1	4	3	2	4	1	3										

SOLUTIONS

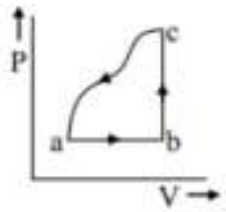
PHYSICS

1)

Heat is exchanged between bodies but not lost to surrounding, so total internal energy of system remains constant.

2) $\frac{dQ}{dt} = -KA\frac{dT}{dx}$ = constant in steady state

$\frac{dT}{dx} \propto \frac{1}{A}$



3)

$\Delta U_{ca} = -180$ J

$\Delta U_{ac} = +180$ J (state function)

$\Delta Q = \Delta U + \Delta W$

$\Delta Q_{a\to c} = \Delta U_{a\to c} + \Delta W_{a\to c}$

$\Delta Q_{a\to b} + \Delta Q_{b\to c} = 180 + \Delta W_{a\to c}$

$250 + 60 = 180 + \Delta W_{a\to c}$

$\Delta W_{a\to c} = 310 - 180 = 130$ J

4)

Solar const. $= \frac{\sigma T^4 R^2}{d^2}$

5)

At const. P

$w = nR\Delta T$

$= 1 \times 8.31 \times (127 - 27)$

$= 831$ J

6) $\gamma = \alpha_x + \alpha_y + \alpha_Z$

$\gamma = \alpha_1 + \alpha_2 + \alpha_2$

$= \alpha_1 + 2\,\alpha_2$

7) By mole conservation

$$\frac{P_1V}{T_1} + \frac{P_2V}{T_2} = \frac{2PV}{T}$$

$$\frac{P}{T} = \frac{P_1}{2T_1} + \frac{P_2}{2T_2}$$

8)

$VT = K \Rightarrow V\left(\frac{PV}{nR}\right) = K$

PV^2 = constant

Comparing with PV^x = constant x = 2

$\Rightarrow C = C_V + \frac{R}{1-x}$

$= \frac{3R}{2} + \frac{R}{1-2} = \frac{R}{2}$

$\Delta Q = nC\Delta T = 1 \times \frac{R}{2}\Delta T = \frac{1}{2}R\Delta T$

9) ice 0°C → water 0°C → water 100°C → steam 100°C → steam 120°C

Q = 10 × 80 + 10 × 1 × 100 + 10 × 540 + 10 × 0.5 × 20

= 800 + 1000 + 5400 + 100

= 7300 cal

10)

At const. P

$\frac{w}{Q} = 1 - \frac{1}{\gamma}$ $\gamma = 1 + \frac{2}{f}$

$\frac{w}{Q} = 1 - \frac{5}{7}$ $\gamma = 1 + \frac{2}{5} = \frac{7}{5}$

$\frac{w}{Q} = \frac{2}{7} \Rightarrow w = \frac{2}{7} \times 140$

$\boxed{w = 40J}$

11) FLOT $Q = \Delta U + W$

$Q = \Delta U + \frac{Q}{5}$

$\frac{4Q}{5} = \Delta U$

$\frac{4}{5}\mu C\Delta T = \mu C_V \Delta T$

$C = \frac{5C_V}{4} = \frac{5}{4} \times \frac{5}{2}R = \frac{25R}{8}$

According to problem $C = \frac{xR}{8} = \frac{25R}{8}$

x = 25

12)

Acc. to weins law

$$\boxed{\lambda_m \propto \frac{1}{T}}$$

13) $ms_1(\Delta T)_1 = ms_2\Delta T_2$

$$\frac{s_1}{s_2} = \frac{\Delta T_2}{\Delta T_1}$$

$$\frac{s}{\left(\frac{s}{2}\right)} = \frac{\Delta T_2}{25}$$

$\Delta T_2 = 50$

14) $V_{rms} \propto \sqrt{T}$

$$\frac{200}{v_2} = \sqrt{\frac{400}{500}} \Rightarrow v_2 = 100\sqrt{5}\ m/s$$

15) $PV = nRT \Rightarrow PV = \frac{m}{M_w}RT$

Slope = $\frac{P}{T} \propto m \Rightarrow \frac{(Slope)_B}{(Slope)_A} = \frac{3m}{m} = \frac{3}{1}$

16) $i = \frac{Q}{t} = \frac{\Delta T}{R}$

$\Rightarrow t \propto R$

$\Rightarrow \frac{4}{t} = \frac{2R}{R/2} \Rightarrow t = 1\ min$

in A thermal resistances are in series whereas in B they are in parallel

17) Heat released by steam to reach 0°C liquid water

$= 10 \times 540 + 10 \times 1 \times 100$

$= 6400$ Cal.

So, ice melted $\Delta m = \frac{6400}{80} = 80$ gm

stem condensed = 10 gm,

total water = 500 + 80 + 10 = 590 gm

18)

For AB, P = const.

For BC, T = const.

For CA, V = const.

19) $mg = m\omega^2 A$

$\omega = \sqrt{\frac{g}{A}} \Rightarrow T = 2\pi\sqrt{\frac{A}{g}}$

20) $A_{resultant} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$

$\Rightarrow \sqrt{3}A = \sqrt{A^2 + A^2 + 2A^2\cos\phi}$

$\Rightarrow 3A^2 = 2A^2 + 2A^2\cos\phi$

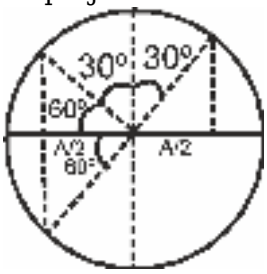
$\Rightarrow \cos\phi = \frac{1}{2}$

$\therefore\ \phi = 60°$

 Phase difference = π/3

21) **Alternate :** by symmetry both should be on opposite side of mean position at equal distance from mean (for max. seperation)

As projection of SHM on circular path



The phase difference from figure is $\frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$

22) $k = \frac{1}{2}m\omega^2A^2\cos^2\omega t$

$U = \frac{1}{2}m\omega^2A^2\sin^2\omega t$

$\frac{k}{U} = \cot^2\omega t = \cot^2\frac{\pi}{90}(210) = \frac{1}{3}$

23) $v = \omega\sqrt{A^2 - x^2}$ .....(1)

$a = -\omega^2 x$ .....(2)

$|v| = |a|$ .....(3)

$\omega\sqrt{A^2 - x^2} = \omega^2 x$

$A^2 - x^2 = \omega^2x^2$

$5^2 - 4^2 = \omega^2(4^2)$

$\Rightarrow 3 = \omega \times 4$

$T = 2\pi/\omega$

24) $x = 3\sin 100t + 8\cos^2 50t$

$= 3\sin 100t + 8\left(\frac{1 + \cos 2\theta}{2}\right)$

$= 3 \sin 100\,t + 4 \cos 100t + 4$
$x = 5 \sin (100t) + 4$
$\Rightarrow A = 5$

25)

$A_1\omega_1 = A_2\omega_2$

$$A_1\sqrt{\frac{k_1}{m}} = A_2\sqrt{\frac{k_2}{m}}$$

26) $T = 2\pi\sqrt{\frac{m}{k}}$

 T is independent of g

27)

V is max. at mean position and mean position will be below natural length.

28) **S-I :** $4 + 6 \sin \omega t$ represents SHM about m.p. = 4.

S-II : $\frac{d^2x}{dt^2}$ is not proportional to x but to (x – 4).

29) $y = 8\sin\left(\frac{\pi x}{20}\right)\cos(50\pi t)$

 Maximum displacement at x = 5 cm is

$$y_0 = 8\sin\left(\frac{\pi \times 5}{20}\right)$$

$$= 8\sin\frac{\pi}{4} = 8 \times \frac{1}{\sqrt{2}} = 4\sqrt{2}$$

$= 5.6$ cm

30)

Given $A_{ref} = = \frac{0.03}{2} = 0.015$

No phase shift at rare medium

So $y = 0.015\sin 8\pi\left[t + \frac{x}{20}\right]$

31)

- Sound intensity decreases with distance as $I \times 1/r^2$
- Decibel scale formula $L = 10 \log_{10}(I/I_0)$

Difference in decibel levels:

$L_1 - L_{min} = 40 - 20 = 20$ dB

Decibel formula:

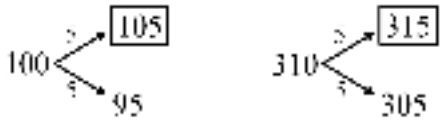
$$L_1 - L_{min} = 10\log_{10}\left(\frac{I_1}{I_{min}}\right)$$

$$20 = 10\log_{10}\left(\frac{I_1}{I_{min}}\right) \Rightarrow \frac{I_1}{I_{min}} = 10^2 = 100$$

Since intensity $I \propto 1/r^2$:

$$\frac{I_1}{I_{min}} = \frac{r_2^2}{r_1^2}$$

$$100 = \frac{r_2^2}{(1)^2} \Rightarrow r_2^2 = 100 \Rightarrow r_2 = 10\text{ m}$$



32) n = 105 Hz 3n = 315 Hz

33) $\Delta n = n_1 - n_2$

$$3 = \frac{v}{2l_1} - \frac{v}{2l_2}$$

$$3 = \frac{v}{2}\left[\frac{1}{50} - \frac{1}{50.5}\right]$$

$$50.5 \times 3 \times 2 \times 50 = 0.5v$$

$$v = 303\,\text{m/s}$$

34)

Beat frequency = $n_2 - n_1$

frequency of amplitude = $\frac{n_2 - n_1}{2}$

frequency of resultant wave = $\frac{n_2 + n_1}{2}$

frequency of intensity variation = $n_2 - n_1$

35) Given $1200 = \sqrt{\frac{\gamma P}{\rho_{H_2}}}$ ...(1)

In mix

$$M_{mix} = M_{H_2} + M_{O_2}$$

If volume of H_2 is x and that of O_2 is y, then

$$(x + y)\,\rho_{mix} = x\rho_{H_2} + y\rho_{O_2}$$

$$\Rightarrow \rho_{mix} = \frac{x\rho_{H_2} + 16y\rho_{H_2}}{(x + y)}$$

So $500 = \sqrt{\frac{\gamma p(x + y)}{(x + 16y)\rho_{H_2}}}$ ...(2)

On $\frac{(1)}{(2)} \Rightarrow \frac{12}{5} = \sqrt{\frac{(x + 16y)}{(x + y)}} \Rightarrow \frac{x}{y} = \frac{256}{119}$

36)

Fundamental frequency of close organ pipe = $\frac{V_1}{4\ell_1}$

Second harmonic frequency of string = $\frac{2V_2}{2\ell_2}$

So $\frac{V_1}{4\ell_1} = \frac{V_2}{\ell_2} = \frac{320}{4 \times 0.8} = \frac{1}{0.5}\sqrt{\frac{50}{\mu}}$

2500 = $\frac{50}{\mu}$

μ = $\frac{1}{50} = \frac{m}{0.5}$

m = 10 gm.

37) $\frac{v}{4\ell_1} = 30 \Rightarrow \ell_1 = \frac{11}{4}m = 275\,cm$

$\frac{v}{4\ell_2} = 33 \Rightarrow \ell_2 = \frac{10}{4}m = 250\,cm$

Length of water = 275 – 250 = 25 cm

Amount of water = ρ × A × □ = 1 × (3 × 25) = 75 g

38) The fundamental frequency of an open organ pipe is n = $\frac{v}{2\ell}$.

As temperature increases, both v and □ increase but v increase more rapidly than □. Hence, the fundamental frequency increases as the temperature increases.

39) Initially f = $\frac{5V}{2\ell} = \frac{5}{2\ell}\sqrt{\frac{16g}{\mu}}$

Now f = $\frac{2v}{2\ell} = \frac{2}{2\ell}\sqrt{\frac{Mg}{\mu}}$

$\Rightarrow \frac{5}{2\ell}\sqrt{\frac{16g}{\mu}} = \frac{2}{2\ell}\sqrt{\frac{Mg}{\mu}}$

M = 100 kg

40) First resonating length = $\frac{\lambda}{4} = \frac{V}{n.4} = \frac{320}{320 \times 4}$

$= \frac{1}{4}m = 25cm$

next resonating length = $\frac{3\lambda}{4} = 75cm$

next resonating length = $\frac{5\lambda}{4} = 125cm$

next resonating length = $\frac{7\lambda}{4} = 175cm$ which is not possible

So max. length of air column at resonance = 125 cm

min. length of water column at resonance = 5 cm
min. length of air column at resonance = 25 cm
max. length of water column at resonance = 105 cm

41) In 1000 s amplitude changes from 8° to 4°
In 2000 s amplitude changes from 4° to 2°
In 3000 s amplitude changes from 2° to 1°
t = 3000 sec = 50 min

42) $5 \times S \times (80–40) = 25\times1\times(40–16)$

$$S = \frac{(25)(24)}{(5)(40)} = 3 cal/gm^oC$$

43) When A & B are mixed together final temp = 12^oC
A = 10^oC, B = 15^oC
$m\times S_A\times(12-10) = m\times S_B\times(15-12)$
$\Rightarrow 2S_A = 3S_B$ .....(1)
When B & C are mixed together, final temp = 21^oC
B = 15^oC , C = 25^oC
$m\times S_B\times(21–15) = m\times S_C\times(25–21)$
$\Rightarrow 6S_B = 4S_C$ .....(1)
$S_A = S_C$
$3S_B = 2S_C$
When A & C are mixed together, final temp = T^oC
A = 10^oC, C = 25^oC
$m\times S_A\times(T–10) = m\times S_C\times(25–T)$
$\Rightarrow T–10 = 25–T$
$\Rightarrow 2T = 35$
$T = 17.5^oC$

44)

Frequency of sonometer wire

$$n = \frac{1}{L}\sqrt{\frac{T}{\rho\pi D^2}}$$

As we know

$$\frac{n_1}{n_2} = \frac{L_2}{L_1}\sqrt{\frac{T_1}{T_2}\times\frac{\rho_2}{\rho_1}\times\left(\frac{D_2}{D_1}\right)^2}$$

Given $\frac{T_1}{T_2} = \frac{8}{1}, \frac{L_1}{L_2} = \frac{36}{35}, \frac{D_1}{D_2} = \frac{4}{1}, \frac{\rho_1}{\rho_2} = \frac{1}{2}$

$$\Rightarrow \frac{n_1}{n_2} = \frac{35}{36}\sqrt{\frac{8}{1}\times\frac{2}{1}\times\left(\frac{1}{4}\right)^2} = \frac{35}{36}$$

According to question, higher frequency = 360

$$\Rightarrow n_2 = \frac{35}{36}\times 360 = 350$$

lower frequency = 350

Beat = 360 − 350 = 10

45) $\lambda = \frac{2\pi}{k} = \frac{2\pi}{15\pi} = \frac{2}{15}$
Shortest possible length
$= \frac{\lambda}{2} = \frac{1}{15}m = \frac{100}{15}cm = 6.66\ cm$

CHEMISTRY

46)

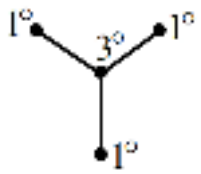
Based on stability of carbocation.

47)

Racemic mixture → Anti elimination → Cis alkene

48)

$CH_3\text{-}C_2\text{-}CH_3 \Rightarrow CH_3 - \overset{sp}{C} \equiv \overset{sp}{C} - CH_3$

49) 

Total 'C' = 4

50) Paper chromatography is a type of partition chromatography.

51) a → r ; Acid and ester are functional isomers
b → p, s ; Ketone shows both position and metamers.
c → r, q ; Functional isomers and tautomer's.
d → r, q Functional isomers and tautomer's.

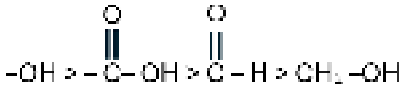
52)

5-Hydroxy cyclohex-3-en-1-one

53) 3D arrangement differ around restricted carbon (double bond)

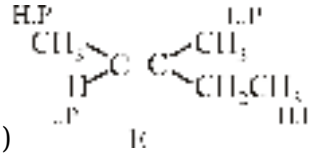
54) **Solution:**

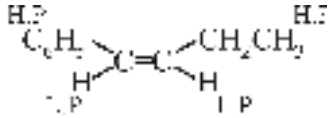
As per the CIP rule
Priority ∝ atomic no of directly attached atom.
∝ If there is tie, then see the atomic no of next connected atoms.
so, correct order of priority is

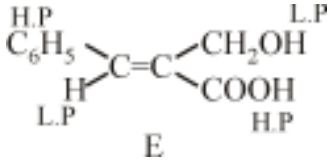


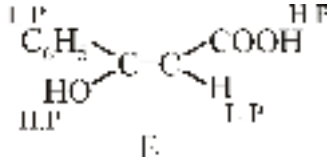
Hence,
Correct option: (2)

55)

(1) 

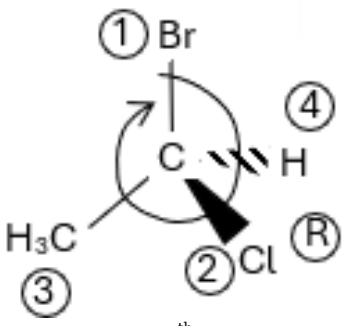
(2) 

(3) 

(4) 

Here : H.P. → High priority group.
L.P. → Low priority group.

56) **Concept:** R/S configuration.
• Follow CIP Rule.
• Give the priority to atoms/group on the basis of CIP Rule
Ans. Exp:-



Least priority (4th) at dash line, so we moves
1 → 2 → 3, we get dockwise direction
So. configuration is R.
Final Answer: (2)

57)

Bond strength ⇒ HF > HCl > HBr > HI
Reactivity ⇒ HF < HCl < HBr < HI

58)

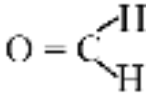
In OMDM cyclic cation forms.

59)

60)

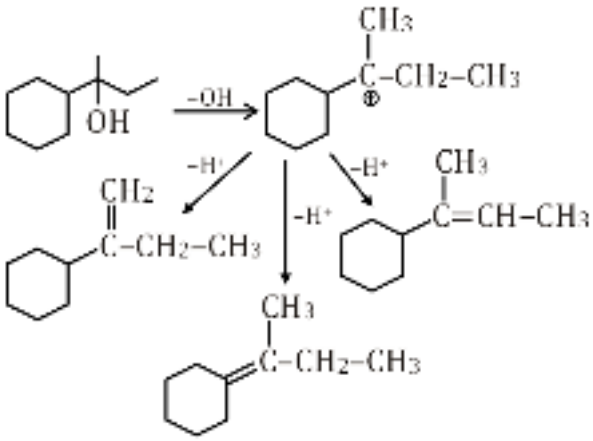
Terminal alkynes give tollen's test. (white ppt)

61) $CH_3 - CH \equiv C \equiv CH_2$

CH_3 – CHO ; CO_2 ; 

62)

In oxidative ozonolysis aldehydes oxidize into –COOH.

63) 

→ So, option (4) is not possible.

64) $R - C \equiv CH \xrightarrow[NH_4OH]{AgNO_3} R–C\equiv CAg \downarrow$
white ppt

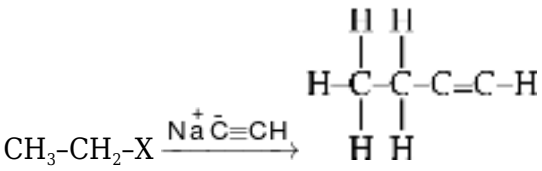
65)

Birch reduction

66)

Markonikovs rule no-2

67)



68)

Rate of FRAR ∝ Stability of free radical.

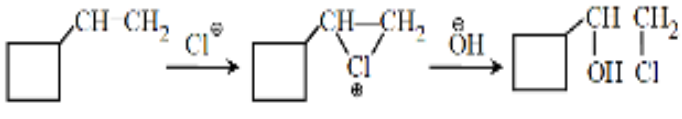
69)

Rate of FRSR ∝ Stability of free radical.

70)

$H_2C = CH_2 \rightarrow$ FRAR
Toluene → FRSR/NBS
Alkane → FRSR

71) Stability of carbocation ∝ $\frac{+M + H + I}{-M - H - I}$

72) 

73) Ease of elimination ∝ (I) stability of Alkene

74) Compounds which follow Huckles rule are aromatic

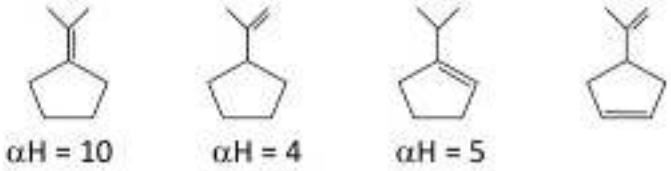
75) In presence of strong Base elimination takes place and reactivity towards eliminant alkyl halides is
R – I > R – Br > R – Cl > R – F

76) β-keto acids are decarboxylated simply on heating.

77) Vacant orbitals containing species are electrophiles

78) Heat of hydrogenation ∝ total no. of d.b.

Heat of hydrogenation $\propto \frac{1}{\text{stability of alkene}}$



79)

$$CH_3 - Cl \xrightarrow[\text{Dry ether}]{Na} CH_3\text{-}CH_3$$

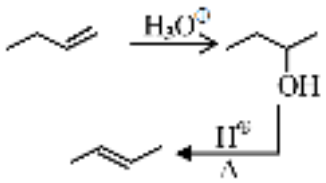
$$CH_3\text{-}CH_3 \xrightarrow[Cl_2/h\nu\downarrow]{} CH_3\text{-}CH_2\text{-}Cl$$

$$CH_3\text{-}CH_3 \xleftarrow[HI]{Red\,P} CH_3\text{-}CH_2\text{-}Cl$$

80)

Reduction by $NH_2\text{–}NH_2/\overset{\ominus}{O}H/\Delta$

81)

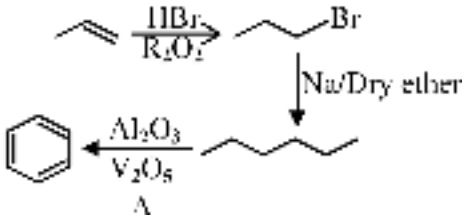


Position ISOMER

82)

A = B = racemic mix.

83)



84)

$$R\text{–}X \xrightarrow[\text{Dry ether}]{Na} R\text{–}R$$

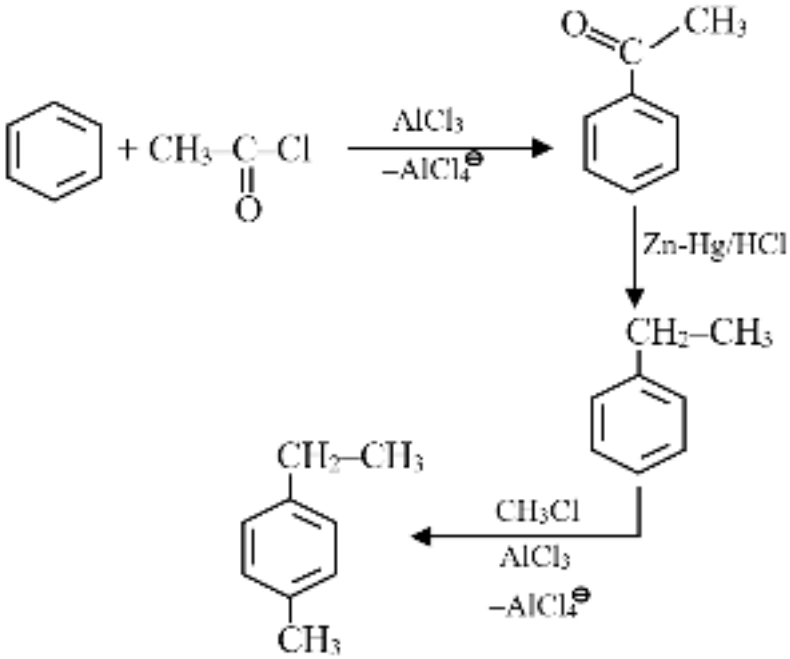
R → Aliphatic → Wurtz

→ Benzene → Fittig

85)

B.P. ∝ Molecular weight ∝ $\frac{1}{\text{Branches}}$

86) **Question Explanation :**
This question is asking about reagent in the given reaction.
Concept :
This question is based on
(i) Friedal-Craft Acylation
(ii) Clemmensen reduction reaction.
(iii) Electrophilic subsititution reaction.
Solution :



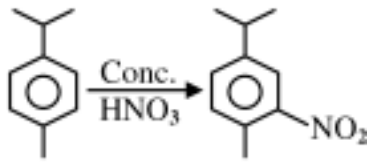
Final Answer : Option (3)

87)

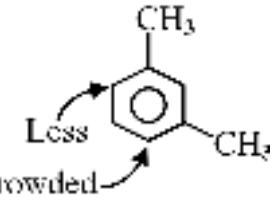
In friedal craft alkylation carbocation is forms.

88)

Rate of ESR ∝ $\frac{+M \ +H \ +I}{-M \ -H \ -I}$

89) 

(more +H)

90) 

BIOLOGY

91)

NCERT XI Pg. # 236,237

92) NCERT Pg. # 230 (Para-14.3)

93)

NCERT XI Pg. # 233

94)

NCERT-XI, Pg. # 240, 243, 244

95) NCERT, Pg # 168

96) NCERT, Pg # 177

97)

NCERT XI Pg # 223

98)

NCERT-XII, Pg # 213

99)

NCERT XI, Pg # 143, 144, 145

100) NCERT XI (E) Page 210, 211 and 213 (2nd para)

101) NCERT XI Pg. # 207, 208

102)

NCERT XI Pg.# 174

103)

NCERT-XI, Pg. # 249 → continued

104) NCERT-XI, Pg. # 229

105) NCERT, Pg # 162

106)

NCERT XI, Pg # 163, 164

107) NCERT-XI Pg#231, PARA-14.4

108)

NCERT XI, Pg # 154

109)

NCERT-XI, Pg. # 229, 234

110)

NCERT XI Pg. # 160

111) NCERT Pg # 236

112) NCERT, Pg. # 232

113) NCERT XI, Pg # 145, 146
In C_4 plants for the production of one glucose = 30 ATP, 12 NADPH and 6 CO_2 same requirement for one fructose.

114)

NCERT (XI) Pg. 213, 215

115)

NCERT XI, Pg # 137, Fig # 11.3a

116) NCERT Pg. # 145

117) NCERT, Pg. # 136

118) NCERT, Pg. 137-138

119) NCERT XI Pg No. # 210

120) NCERT, Pg. # 134

121) NCERT, Pg. # 135

122)

NCERT Pg. # 141

123)

NCERT (XII)th Pg. # 139,140,141,142

124)

NCERT XI, Pg # 145

125) NCERT XII Page No. # 173

126) NCERT XI, Pg. # 172

127) NCERT Pg.,# No. : 177

128) NCERT Pg. # 175

129)

NCERT XI, Pg # 172

130) NCERT, Pg. # 176,177

131) NCERT (XI) Pg. No. # 160

132) NCERT Pg. # 158

133) NCERT XI, Pg. # 158

134) NCERT Pg. # 171

135)

NCERT XI, Pg # 174, 175

136) NCERT Page No. 236

137)

NCERT Page No.# 231

138) NCERT-XI Page No. 234

139) NCERT-XI Page No. 232

140) NCERT Pg. No. # 321

141) NCERT XI, Pg. # 236

142) NCERT Pg. # 235

143) NCERT-XI Page No. 231

144)

NCERT-XI Page No. 232, 233

145) NCERT-XII Page No. 232

146)

NCERT-XI Page No. 232

147) NCERT Pg.#236.

148) NCERT Pg.#232,236,235.

149) NCERT, Pg. No. 236

150)

NCERT Page No.# 243

151) NCERT, Pg # 239, 240

152) NCERT, Pg. # 248

153) NCERT, Pg. # 248

154) NCERT, Pg. # 242, 243, 246

155) NCERT, Pg. # 245 to 247

156) NCERT Pg # 248

157) NCERT Pg # 246

158) NCERT Pg.#239.

159) NCERT Pg.#242.

160) NCERT Pg.#250

161) NCERT, Pg. # 242, 243, 244

162) NCERT, Pg. # 248, 249

163) NCERT XIIth Pg. # 242-243

164) NCERT, Pg. # 243

165) NCERT Pg # 247

166) NCERT Pg # 246

167)

NCERT Page No.# 231

168) NCERT Pg. # 222

169) NCERT Pg. # 218

170) NCERT Pg.#219.

171) NCERT Pg.#217,218.

172)

Question Explanation : The question is asking about the changes in lengths of different regions of a sarcomere.
Concept : Sarcomere
Solution :
B is I-band, which shortens as actin filaments slide over myosin.
C is H-zone, which also shortens as thick and thin filaments overlap more.
A-band remains constant because the length of thick filaments does not change.
Final Answer : Option (3)

173) NCERT- XI, Pg.# 218

174) NCERT Pg.#219,228

175) NCERT Pg # 227

176) NCERT Pg # 227

177)

NCERT Page No.# 227

178) NCERT Pg # 225

179) NCERT Pg.#227

180) NCERT, Pg. # 227